From EMBER to CINDER: Sharpen the Concept, Burn It Once in Eight Languages, Spare the Word

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Abstract

Embedding-level concept erasure removes a concept from a language model by editing its token embeddings, but it can only remove the concept features it finds. We show that EMBER often misses them: concept tokens are rare, and rare tokens share a common direction that groups the concept with unrelated words. We introduce CINDER (Contrastive INformed Directional Embedding Removal), which sharpens the embeddings toward the concept against ordinary text, lets each feature take as many tokens as the data supports, and verifies features against a lexicon generated once per concept. The same sharpening extends embedding-level erasure, so far limited to English, to eight languages: a single fit with the uncentred CINDER-z edit comes within 0.02 of per-language EMBER at an eighth of the judge calls. We further show that the standard evaluation rewards edits that destroy the edited words, and propose a collateral-aware evaluation that charges for this damage. Under it, CINDER erases concepts that EMBER misses and, at moderate edit size, removes more, and embedding erasure is revealed to block access to a concept through its words rather than remove it from the weights.

1 Introduction

As language models (LMs) are deployed more widely, erasing specific knowledge from them becomes important for safety and compliance (Eldan and Russinovich, 2023; Li et al., 2024). *Persistent* concept erasure does so by modifying the model's weights (Gur-Arieh et al., 2025; Ashuach et al., 2026), and Suslik et al. (2026) showed that such methods overlook a part of the model that stores knowledge: the token embeddings. Their method, EMBER, decomposes the embeddings of concept-related tokens into sparse features, selects the concept's features with an LLM judge, and subtracts them from a small set of tokens; combined with MLP-based methods, it substantially improves robustness to relearning.[1]

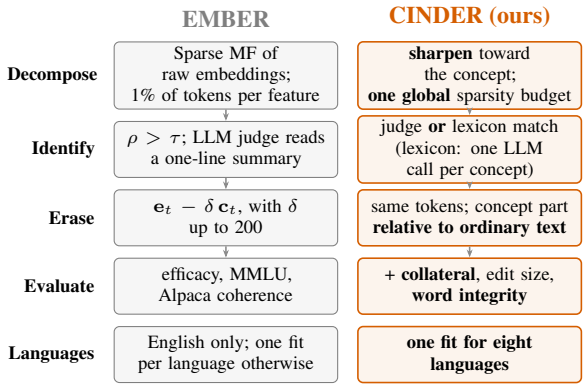


Figure 1: EMBER decomposes token embeddings into sparse features, identifies the concept's features and subtracts them from concept tokens. CINDER keeps this framework and changes how the concept is found, verified and edited; one CINDER fit serves eight languages. Our evaluation no longer counts destroying the edited words as erasing the concept.

However, an embedding edit can only remove what its features capture, and how reliably EMBER finds them has not been studied (Suslik et al., 2026, Limitations). We hypothesize that the problem lies partly in the geometry EMBER searches: concept tokens are rare, and rare tokens share a large component along the mean embedding of ordinary text, so EMBER groups the concept with other rare tokens: its Harry Potter feature also collects Nestlé, Lego and Trump. Whitened, the concept forms a clean but *quiet* cluster, of ordinary value to a factorization that minimizes reconstruction error (Figure 2). A fixed feature width and a judge that reads only a one-sentence summary lose more, and when no feature is accepted, nothing is erased.

We introduce CINDER (Contrastive INformed

[1]Our code is available at https://github.com/guretskiysemyon/nlp_proj.

Directional Embedding Removal; Figure 1), which keeps EMBER's localize-then-edit framework and changes how the concept is found. It whitens the embeddings against ordinary text, removing the shared component, and sharpens them toward the concept's page, making the clean cluster loud; it lets each feature take as many tokens as the data supports; and it verifies features both bottom-up, with EMBER's judge, and top-down, against a lexicon generated once per concept. Its edit removes only what distinguishes the concept from ordinary text, while the variant CINDER-z, like EMBER, also removes part of what all tokens share.

Sharpening also makes multilingual erasure practical, which EMBER leaves to future work. EMBER edits English token rows, so the concept survives in other languages; running it per language multiplies its cost by the number of languages, and a single EMBER fit on multilingual text accepts a concept feature for only 10 of 38 concepts, mostly because its judge rejects the concept features the fit finds. CINDER whitens against a background balanced across languages, sharpens toward the concept's pages in all of them and adds its lexicon judge, and on Llama-3.1-8B-Instruct (Grattafiori et al., 2024) a single CINDER-z fit over eight languages comes within 0.02 of eight separate EMBER runs, with an eighth of the judge calls: to our knowledge, the first multilingual erasure at the embedding level.

Evaluating CINDER on Gemma-2-2B-it (Gemma Team et al., 2024) exposed a blind spot in the standard protocol: EMBER's tuning score prefers edits up to a hundred times longer than the embeddings they change, which destroy the edited words, and since the concept questions contain these words, a destroyed word counts as erased knowledge. We add *collateral* questions that use the edited words in other meanings (e.g., Prince Harry when erasing Harry Potter), a common scale of edit size, and tests of whether the edited words survive. Once collateral damage is charged for, every method moves to edits about as long as the token, and embedding erasure turns out to block access to a concept through its words rather than remove it from the weights. In this regime, CINDER erases concepts that EMBER misses (in 14 runs only CINDER finds the concept, in 3 only EMBER) and, at a moderate equal edit size, removes more ($+0.15$ on the combined score, 10 of 11 held-out concepts); at each method's own operating point, the two tie.

To conclude, our work (1) diagnoses why embedding-level erasure misses quiet concepts; (2) introduces CINDER, which finds and erases them more reliably, with a judge that checks each feature both bottom-up and top-down; (3) extends embedding-level erasure to eight languages with a single fit and an eighth of the judge calls; and (4) shows that the standard evaluation rewards destroying tokens, and proposes a collateral-aware alternative with no LLM judge in the tuning loop.

2 Preliminaries and Notation

We follow the notation of Suslik et al. (2026): a transformer LM (Vaswani et al., 2017) has hidden dimension d, vocabulary $\mathcal{V}$ and input embedding matrix $E \in \mathbb{R}^{|\mathcal{V}|\times d}$, whose row $\mathbf{e}_t \in \mathbb{R}^d$ embeds token t.

Concept Erasure in Language Models Given a model M and a target concept C (e.g., Harry Potter), persistent concept erasure updates the parameters of M so that it no longer answers questions about C better than chance, while maintaining its performance on other concepts and tasks (Eldan and Russinovich, 2023; Li et al., 2024; Gur-Arieh et al., 2025; Suslik et al., 2026). We assume target sentences $\mathcal{S}_C$ from C's Wikipedia page and neutral sentences $\mathcal{S}_N$ from unrelated pages.

EMBER EMBER follows a localize-then-edit approach with three stages. (1) *Decomposing.* Let $\mathcal{V}^*$ be the token types that appear in $\mathcal{S}_C \cup \mathcal{S}_N$, $n = |\mathcal{V}^*|$, and $A = E_{\mathcal{V}^*}^\top \in \mathbb{R}^{d\times n}$. EMBER factorizes $A \approx ZY$, with a dictionary $Z \in \mathbb{R}^{d\times k}$ of *embedding features* $\mathbf{z}_i$ and coefficients $Y \in \mathbb{R}^{k\times n}$, keeping the 1% largest entries of each row of Y and alternating least squares (Sparse MF). (2) *Identifying.* Tokens are labelled concept, neutral or both, by whether they appear in $\mathcal{S}_C$, $\mathcal{S}_N$ or both; with T_C and T_N the concept- and neutral-labelled tokens, each feature is scored by its mass ratio

$$\rho_i = \frac{\sum_{t\in T_C} |Y_{i,t}|}{\sum_{t\in T_N} |Y_{i,t}|}, \tag{1}$$

and features with $\rho_i > \tau = 2$ go to an LLM judge that first describes the feature's top tokens without seeing the concept, then decides from the description alone whether the feature is the concept, accepting at confidence ≥ 0.85. The accepted features form $\mathcal{F}_C$. (3) *Erasing.* Each concept token

with $Y_{i,t} \neq 0$ for some $i \in \mathcal{F}_C$ is edited as

$$\mathbf{e}'_t = \mathbf{e}_t - \delta\,\mathbf{c}_t, \qquad \mathbf{c}_t = \sum_{i \in \mathcal{F}_C} Y_{i,t}\,\mathbf{z}_i, \quad (2)$$

where $\mathbf{c}_t$ is the token's *concept part* and $\delta \geq 0$ the erasure strength.

Evaluating Erasure EMBER measures efficacy (accuracy on multiple-choice questions about C), specificity (similar-domain questions and MMLU; Hendrycks et al., 2021), coherence (AlpacaEval judged by an LLM; Taori et al., 2023) and robustness to relearning (Deeb and Roger, 2024); the harmonic mean H of all but relearning, normalized by the unedited model, selects $\delta \in \{0.5, \ldots, 200\}$ on validation questions.

3 Why EMBER Misses Quiet Concepts

How reliably does EMBER find the target concept? We reproduced it on Gemma-2-2B-it, matching the published accuracies within four points (§A), inspected each stage on its flagship concept, Harry Potter, and tested most observations on 38 concepts and five seeds. We find five issues, which Figure 1 maps to CINDER's changes.

The concept hides behind rarity Sparse MF minimizes squared reconstruction error, so a feature is worth the energy it explains. Concept tokens carry as much energy as neutral ones but are rarer (in 35 of 38 concepts), and in Gemma's embeddings rare tokens share a large component along the background mean μ_B, the average embedding of ordinary text (Figure 2a; cf. Gao et al., 2019). EMBER's concept feature is therefore loud for the wrong reason: 42% of its squared norm lies along $\hat{\mu}_B = \mu_B/\|\mu_B\|$, against 12% for a typical feature, and it groups the concept with unrelated rare tokens, which EMBER then edits (§I). Whitening removes the shared component but leaves a pure, ordinary cluster, which the factorization finds less often than before (Figure 2b): in this sense the concept is *quiet*.

One width does not fit all concepts EMBER keeps the top 1% of tokens, about 48, in every feature. Ranked by their share of the concept feature, tokens stay specific to the concept's page up to a point we call the *natural tail*; it comes after 48 for 15 of 38 concepts, which EMBER cuts short, and before 48 for another 15, whose features it pads with nonspecific tokens (§C.1).

The judge loses the concept in its summary The judge decides from a one-sentence description, and the description is where concepts get lost: Heroin and World War II features became "substances, medications, and illicit drugs [...]" and "20th-century totalitarian political ideologies [...]", and the judge ruled them not to be the concept (§I). A rejected concept is not erased at all; on 12 held-out concepts, EMBER edits nothing in 19 of 60 runs.

The edit stays in English On Gemma, EMBER-edited models retain 0.19–0.24 more of the concept's knowledge in German, Spanish, French and Russian than in English (§F). On Llama-3.1-8B, a single EMBER fit on text in all eight languages accepts a concept feature for only 10 of 38 concepts, mostly because its judge rejects the concept features the fit finds (§5.3). In such a fit, the concept also competes with language: what separates the eight languages carries 34 times the energy of the concept (§F).

The evaluation rewards destroying tokens H selects the largest strength, $\delta = 200$, for 6 of the 10 concepts EMBER edits in our reproduction. The edited embedding is then 40–120 times longer than the original and points along $-\mathbf{c}_t$: the word is replaced rather than cleaned. No metric registers this: the concept questions contain the edited words, so a destroyed word counts as erased knowledge, and the other checks rarely use them (§A). Prompts that use these words in other senses do break, 12 of 20 on Harry Potter at $\delta = 200$ against 0–1 at $\delta = 1$ (Table 3): we call this *token destruction*.

4 CINDER

CINDER (Contrastive INformed Directional Embedding Removal) keeps EMBER's data, Sparse MF, ρ filter and edit form (Eq. 2), and changes how features are decomposed (§4.1), identified (§4.2) and erased (§4.3); §4.4 extends it to many languages with one fit.

4.1 Finding Concept Features in a Sharpened Space

Sparse MF finds the directions with the most energy; rather than changing the factorization, we change the matrix so that the concept carries that energy, combining whitening, a standard step before sparse coding (Hyvärinen and Oja, 2000),

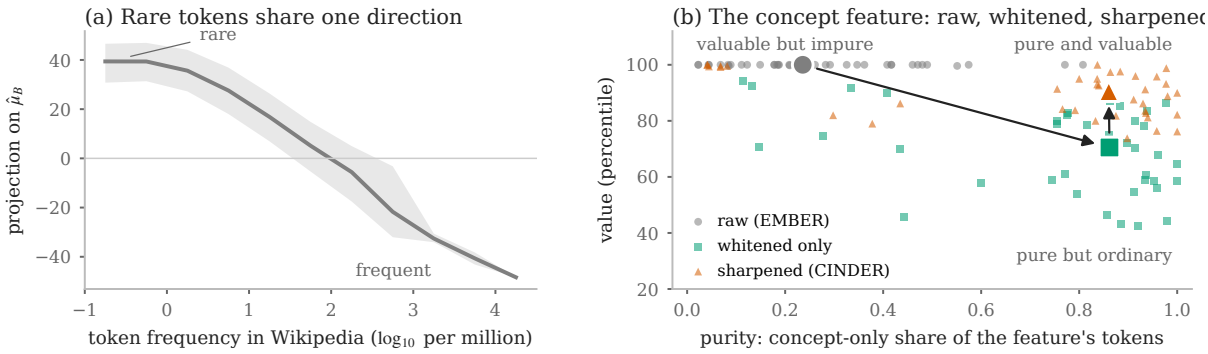


Figure 2: Why EMBER misses quiet concepts (Gemma-2-2B-it). (a) Rarer tokens share a larger component along the background mean $\hat{\mu}_B$ (median and interquartile range per frequency bin). (b) A sparse feature started on each concept's lexicon tokens and refitted until its tokens stop changing, per concept (small markers) with medians (large): in the raw matrix it is valuable but drifts to other rare tokens; after whitening it is pure but ordinary; after sharpening, pure and valuable.

with a target–background contrast (Abid et al., 2018).

Contrastive Sharpening Contrastive Sharpening (CS) *whitens* the embeddings against ordinary text, with the mean μ_B and covariance Σ_B of the token types of a large Wikipedia sample (computed once per model), and *sharpens* them toward the whitened token types $\tilde{\mathbf{u}}_1, \ldots, \tilde{\mathbf{u}}_m$ of the concept's Wikipedia page:

$$\tilde{\mathbf{u}}_t = W(\mathbf{e}_t - \mu_B), \quad W = (\Sigma_B + \varepsilon I)^{-1/2},$$
$$\mathbf{x}_t = S_\eta^{\gamma/2}\,\tilde{\mathbf{u}}_t, \quad S_\eta = (1-\eta)\,\tfrac{1}{m}\textstyle\sum_j \tilde{\mathbf{u}}_j\tilde{\mathbf{u}}_j^\top + \eta I, \tag{3}$$

with $\eta = 0.05$ and $\gamma = 1$; Sparse MF then factorizes $X = T(A - \mu_B \mathbf{1}^\top) \approx \tilde{Z}Y$ with $T = S_\eta^{\gamma/2} W$. After whitening, the page's second moment along a direction measures how much more the concept uses it than ordinary text does, and sharpening multiplies the energy along each such direction by this ratio raised to γ, as in Fisher's discriminant and contrastive PCA (Fisher, 1936; Abid et al., 2018) (Proposition 1, §B). Whitening removes the component that rare tokens share, so the concept's tokens separate from other rare tokens, but makes every direction equally strong; sharpening then makes the clean cluster worth a feature (§6). Since T is invertible, the factorization maps back exactly: $\mathbf{c}_t = T^{-1}\tilde{\mathbf{c}}_t$.

Global Sparsity EMBER's row sparsity (RS) bounds every feature separately, $\|\mathbf{y}_i\|_0 \le s\,n$; Global Sparsity (GS) bounds the whole matrix with the same total budget, $\|Y\|_0 \le s\,k\,n$. After each dense update of Y, GS keeps the $s\,k\,n$ entries with the largest share of their row, $|y_{ij}|/\|\mathbf{y}_i\|_2$, with a floor of 0.2% and a cap of 3% of the tokens per feature. For this score, it is the exact projection onto the constraint, as in iterative hard thresholding (Blumensath and Davies, 2009), and mirrors across features what BatchTopK does across examples (Bussmann et al., 2024); the scale-free score keeps a few high-energy generic features from taking the budget (§C).

4.2 Identifying Concept Features from Both Directions

EMBER's token judge (TJ) works bottom-up, abstracting a feature into a sentence, and fails when the abstraction drops the concept. Inspired by query expansion (Wang et al., 2023; Gao et al., 2023), the Two-Way Judge (TWJ) adds a top-down path: once per concept, before any feature is judged, an LLM expands the concept into a lexicon L_C of 20 words specific to it rather than to its broader category. Each feature's top tokens are mapped to the words they come from in $\mathcal{S}_C$ (e.g., *ffindor* $\rightarrow$ *Gryffindor*), giving a word list W_i, and the expansion judge accepts the feature if $|W_i \cap L_C| \ge t$ (EJ@t); TWJ accepts if either path does (TJ $\vee$ EJ@5). The union raises recall, t controls precision, and the lexicon path adds no LLM call per feature (§D).

4.3 Erasing With and Without the Mean

For fixed coefficients, the least-squares dictionary does not depend on the metric in which the error is measured (Proposition 2, §E), so whitening does not distort the mapped-back edit; what differs from EMBER is the centring. CINDER's concept part describes how a token departs from ordi-

nary text, whereas EMBER's features also absorb a share of μ_B, the component that all tokens share (44% of its edit's energy on Gemma), which the model's first layer reads strongly (§6). CINDER-z keeps the sharpened fit's coefficients Y, selected features and edited tokens, and re-solves only the dictionary on the raw embeddings,

$$Z_z = A\,Y^\top \big(YY^\top + \lambda I\big)^{-1}, \qquad (4)$$

a $k \times k$ solve with no new fit or judge call that adds exactly a mean term, $\mathbf{c}_t^z = \mathbf{c}_t + \beta_t\,\mu_B$ (Proposition 3).

4.4 One Fit for Eight Languages

CINDER erases a concept in eight languages with one fit. Because the concept competes with language identity, which multilingual models encode in a few dominant directions (Chang et al., 2022; Libovický et al., 2020), CINDER whitens against a background with equal text in every language and sharpens toward the concept's pages in all of them (§B.4). It factorizes only a sample, 30 concept and 30 neutral sentences per language, to obtain the dictionary Z, and with Z fixed solves the coefficients Y of every token in the full sentence sets (up to 300 concept and 300 neutral per language). TWJ then selects features once for all languages, and with Y fixed, Z is re-solved on the unsharpened embeddings to give the CINDER-z edit (Eq. 4). Against eight per-language EMBER runs, this takes a fifth of the factorization work, as it factorizes a fifth of the rows, and an eighth of the judge calls, but a third of the LLM cost, as the eight-language lexicon is a long call; fit time stays about the same, because the sharpening transform itself takes time, and evaluation, most of the compute, is the same for all approaches (§F).

5 Experiments

We test whether the standard protocol can tell erasing a concept from destroying its words, how CINDER compares with EMBER under a collateral-aware evaluation, and whether one CINDER fit can replace per-language EMBER.

5.1 Experimental Setup

Models and Data We use Gemma-2-2B-it (Gemma Team et al., 2024) in English and Llama-3.1-8B-Instruct (Grattafiori et al., 2024) in eight languages. As in EMBER, $\mathcal{S}_C$ and $\mathcal{S}_N$ hold 300 Wikipedia sentences each, with 50 validation and 50 test questions per concept. We developed the evaluation on Harry Potter, Ancient Rome and Valentine's Day and test on 12 of EMBER's other concepts with 5 seeds each (60 runs); the multilingual study uses EMBER's 18 concepts and 20 new ones held out under a pre-registered protocol (§F).

Evaluation Metrics We keep EMBER's metrics except relearning (§2) and add two that target the edited tokens; the multilingual study keeps EMBER's alone, with perplexity on native text in place of Alpaca. *Collateral* is accuracy on multiple-choice questions that use an edited token outside the concept, such as *Harry* in Prince Harry or *Caesar* in Caesar salad (3,501 items, written by LLM agents and frozen before scoring). *Word integrity* tests whether the model can still copy, define and read past an edited word, and keep its answers on neutral prompts that use it, against two references: the word deleted, and a *nonce*, a pronounceable non-word that the model can repeat but does not understand, which is how an erased word should behave. Operating points are selected by $H_3 = \mathrm{HM}(\text{efficacy}, \text{specificity}, \text{collateral})$ among points that keep MMLU and collateral above 0.9 and 0.8 of the unedited model. Since equal δ does not mean equal edits across methods, all methods sweep one grid of *edit sizes* $r = \mathrm{median}_t \,\|\mathbf{e}'_t - \mathbf{e}_t\|/\|\mathbf{e}_t\| \in \{0.05, \ldots, 100\}$, where $r = 1$ means that each edit is as long as the token. Scoring a grid point needs only forward passes, with no LLM judge (§G).

Methods and Statistics We compare EMBER, CINDER (CS + GS + TWJ), CINDER-z and, to isolate Global Sparsity (GS), CINDER-RS and CINDER-RS-z with row sparsity; in the multilingual study, we compare EMBER run per language with single fits of EMBER, CINDER and CINDER-z over all eight languages (A1–A3z in §F). Each method selects one operating point per run on validation questions (besides our rule, we report EMBER's H without Alpaca and a fixed $r = 0.5$) and is scored once on test questions. Comparisons are paired by run, with the concept as the unit (Wilcoxon tests, Holm correction, bootstrap 95% intervals); only the comparisons under our rule, H and fixed r and the multilingual hypotheses (§F) were pre-registered, and other analyses are post hoc unless marked.

Method, rule	r	Eff.↑	Coll.↑	Broken↓	Alp.↑
EMBER, H	100	0.47	0.73	0.24	0.97
EMBER, H_3	1	0.29	0.85	0.15	0.97
CINDER-z, H	100	0.68	0.68	0.34	0.97
CINDER-z, H_3	1	0.39	0.86	0.16	0.98

Table 1: Points selected by EMBER's H (without Alpaca) and our H_3 on 12 held-out concepts (test). Broken: neutral prompts whose answer breaks (9 concepts, one seed); Alp.: Alpaca coherence (seed 42).

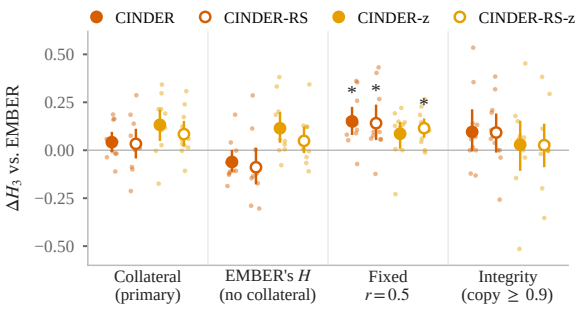


Figure 3: Paired test H_3 difference to EMBER on the 11 held-out concepts where both edit, by selection rule (integrity-gated: copy $\geq$ 0.9, post hoc). Dots: concepts; bars: 95% bootstrap CI; $*$: Holm $p < 0.05$.

	EMBER per lang.	EMBER one fit	CINDER one fit	CINDER-z one fit
Judge calls	160	20	20	20
LLM cost (USD)	.015	.0019	.0047	.0047
Concepts edited	26	10	21	21
Score, strict	0.25	0.11	0.14	0.23
Score, ungated	0.25	0.14	0.14	0.28

Table 2: Multilingual erasure on Llama-3.1-8B-Instruct (38 concepts, 8 languages, fit seed 42, test split): per-language EMBER against single fits over all eight languages. Calls and cost are per concept; concepts without an edit score 0.

5.2 Results

The standard protocol rewards token destruction Without a collateral term, selection settles on edits a hundred times the token's length for EMBER and CINDER-z (Table 1); with it, every method moves to $r \approx 1$. At the destructive points, 24–34% of neutral prompts break, and efficacy exceeds the level reached by deleting the edited words from the questions (0.23–0.28) in 83% and 98% of runs: the edits likely derail the prompt as well as delete the word. Alpaca coherence registers none of this, and under H the ranking reappears through destruction, with EMBER ahead of CINDER (n.s.).

CINDER erases concepts that EMBER misses EMBER edits in 41 of the 60 held-out runs and CINDER in 52; in runs where only one of them edits, CINDER does so in 14 and EMBER in 3 (run-level exact McNemar $p = 0.013$, post hoc).

At moderate edit size, CINDER's direction removes more At $r = 0.5$, every CINDER variant removes more than EMBER at the same collateral (efficacy 0.25–0.30 vs. 0.20); for CINDER, the gain is $+0.15$ H_3 on 10 of 11 concepts (Holm $p = 0.012$; Figure 3). Crossing the two token sets with the two directions (post hoc) attributes it to the direction ($+0.07$ efficacy, 10 of 11 concepts, Holm $p = 0.10$) rather than the token set ($+0.02$; §G.6). At $r = 1$ it vanishes (-0.02, $p = 0.97$; post hoc): on the development concepts, centred directions plateau near the deletion floor by then, and EMBER passes it only by damaging the prompt (§E.1).

At their own operating points, the methods tie When each method picks its own strength, the preregistered primary comparison shows no significant difference ($+0.04$, Holm $p = 0.41$): both settle near $r = 1$, where the direction effect vanishes. With a stricter collateral gate (0.9), all four CINDER variants beat EMBER significantly, though not with a looser one (0.7; post hoc; §G).

5.3 Multilingual Erasure

One CINDER fit replaces eight EMBER fits Counting concepts without an edit as failures, the single CINDER-z fit ties per-language EMBER (exploratory; non-inferiority not shown): 0.23 vs. 0.25 strict and 0.28 vs. 0.25 ungated, 15–13 in wins and losses ($p = 0.78$; Table 2).

A multilingual EMBER fit rarely accepts the concept Where the multilingual EMBER fit selects a concept feature, it erases as well as CINDER-z, but it does so for only 10 of 38 concepts and 2 of the 20 held-out ones, against 21 and 10 for CINDER; counting coverage, CINDER-z beats it by $+0.21$ (15–4, $p = 0.011$). The loss is mostly at the judge: the raw fit contains a lexicon-matching concept feature about as often as CINDER's, but EMBER's judge accepts few of them; re-judged with TWJ, the raw fit reaches 18 concepts instead of 10, and sharpening alone 16 (seed 42, post hoc; §F).

The mean term matters across languages Multilingual CINDER, whose edit is centred, erases weakly; its uncentred CINDER-z edit (defined after the development results) adds $+0.17$ (16 of the 21 edited concepts, $p = 0.007$), in the same direction on the 20 pre-registered held-out concepts ($+0.19$, 7–3, $p = 0.064$). Across all approaches, erasure is strongest in English, about 0.6 of that in the other Latin-script languages, little in Thai and almost none in Hindi, whose words split into pieces that are rarely concept-only (§F).

6 Embedding-Space Analysis

Why does CINDER find concepts that EMBER misses? We analyze the features before any edit (38 concepts, 15 added after the method was frozen; 5 seeds), judge-free wherever possible (§H).

Sharpening makes the concept pure and valuable We start a single sparse feature on each concept's lexicon tokens and repeat EMBER's fitting step until its tokens stop changing (Figure 2b; §B.2). In the raw matrix it drifts to other rare tokens, ending only 24% concept-only yet among the most valuable features; whitening makes it pure (86%) but ordinary (70th percentile of feature energies), and sharpening keeps it pure and makes it valuable (91st; above whitening in 37 of 38 concepts). Both are needed in ordinary fits, where 100 features compete: a lexicon-matching concept feature emerges in 69% of sharpened runs, against 56% raw and 49% whitened-only. The raw feature's value is mostly the shared mean ($\hat{\mu}_B$ holds 97% of its energy above an average feature's), which also makes its nearest neighbour a feature of unrelated rare names; after sharpening, the neighbour is a related topic (magic for Harry Potter; §H.2).

Widths follow the data On the sharpened matrix, EMBER's 1% cut stops before the natural tail (§C.1) for 25 of 38 concepts, and Global Sparsity (GS) widths follow the tail (rank correlation 0.67 vs. 0.27 for the fixed width; mean error 30 vs. 42 tokens, $p < 0.001$), though they compress its range; GS changes which tokens a feature keeps, not its direction (cosine 0.97), and helps only after Contrastive Sharpening (CS).

The two-way judge recovers concepts lost in the summary On 19 concepts, TWJ accepts the concept or a broader version of it in 82 of 91 runs, against 61 for EMBER's judge, with 1 of 110 acceptances unrelated, though most added features are broader than the concept. On 15 harder concepts added later, this gain mostly disappears (40 vs. 34 of 75 runs), but a second pass in which the judge also sees each token's source word recovers it (52; §D).

The gain is not circular The concept's page, our sharpening target, contains the sentences whose tokens we score, yet sharpening toward page sentences outside the sample or toward other pages about the concept raises the strict purity of the concept feature as much ($+0.06$ vs. $+0.05$; Holm $p = 0.034$, 0.081; pre-registered), while another concept's page destroys it (found in 2 of 54 runs; §B.3).

Abstract concepts stay out of reach On 13 abstract concepts such as Stoicism and Gothic architecture, the top-ranked feature is specific to the concept in 0 of 65 runs under both EMBER and CS, against 13 and 23 of 40 for entity concepts; their top features are parent topics and discourse words, perhaps a register they share (untested).

The first layer reads the shared mean The first layer responds to EMBER's and CINDER-z's edits about twice as strongly as to a random direction, and to CINDER's 1.2× (Figure 4, §E). The gap is the mean term: about half of an uncentred edit lies along $\hat{\mu}_B$, which the first layer amplifies (2.9× on Gemma, 20× on Llama); without it, the two edits draw the same response (1.31× vs. 1.32×).

7 Token-Level Analysis: What an Embedding Edit Removes

Does an embedding edit remove the concept from a word, or the word from the model? Since the input embedding is where the model first reads a token, we expect an edit to control *access* to the concept through the edited words, at a cost to the words.

Embedding erasure blocks access Since only input rows change, efficacy questions that contain none of the edited tokens are answered exactly as before (efficacy 0.000 in all 22 configurations checked, including $r = 100$): the knowledge remains in the weights, cut off from the edited words. At the points our rule selects ($r \approx 1$), efficacy lies close to the floor reached by deleting those words

Prompt	$\delta = 1$ ($r \approx 0.6$)	$\delta = 200$ (selected by H)
Who is Prince **Harry**, and who is his wife?	*Prince Harry is the second son of King Charles III [. . .] His wife is Meghan Markle*	*Prince was an American singer, songwriter [. . .] Prince's wife was Mayte*
What country does the toy company **Lego** come from [. . .]?	*Lego comes from Denmark.*	*Please provide me with the name of the toy company!*

Table 3: Gemma-2-2B-it after erasing Harry Potter with EMBER (seed 42) at a moderate strength and at the strength EMBER's score selects. Edited tokens, both from the Harry Potter feature, are in bold.

from the questions, above it only for the -z variants.

Removing more costs the words At the selected points, about half of the edited words can no longer be read (0.49–0.61, seed 42), and most of the concept-exclusive ones (0.65–0.81), with no significant difference between methods; Table 3 shows destruction in free text. Requiring the words to stay copyable (post hoc) caps efficacy at 0.06–0.14; under this constraint CINDER removes about twice as much as EMBER (n.s.), whereas CINDER-z's advantage disappears. On the development concepts, its mean term alone erases more than CINDER-z at $r \geq 2$ by breaking the prompt (an unrelated question after the word falls below chance), and almost nothing when the words must stay copyable (§E.1).

A nonce-like edit is only partly reachable An erased word should behave like a nonce, copyable but meaningless; a learned edit that pushes each edited word toward its nonce's behaviour (post hoc) removes 2–3× as much as copy-gated EMBER while keeping the words copyable, but reaches a substantial nonce floor in only 2 of 12 concepts (seed 42; §G.7). Figure 6 (§G) traces this trade-off along the edit-size grid.

8 Related Work

Concept Erasure and Its Evaluation Unlearning methods optimize a forget objective (Eldan and Russinovich, 2023; Li et al., 2024; Ashuach et al., 2026) or edit the parameters that store the concept (Gur-Arieh et al., 2025; Shafran et al., 2026; Suslik et al., 2026); in both, erasure is often undone by relearning (Deeb and Roger, 2024), adversarial prompts (Lynch et al., 2024) or another format (Ye et al., 2025; Hong et al., 2025). Knowledge editing tests specificity on *neighborhood* prompts that share surface form with the edit (Meng et al., 2022); our collateral term brings this idea to embedding erasure, and our nonce reference follows the *wug* test (Berko, 1958).

Sparse Dictionaries and Their Interpretation Sparse factorizations recover interpretable parts of data (Lee and Seung, 1999; Hoyer, 2004) and of model activations (Bricken et al., 2023; Huben et al., 2024; Gao et al., 2025), with the sparsity allocation shaping what is found (Makhzani and Frey, 2014; Bussmann et al., 2024); their features are commonly named by an LLM (Bills et al., 2023; Paulo et al., 2024), whose verbalized confidence is poorly calibrated (Xiong et al., 2024).

Geometry of Token Embeddings Embedding spaces are anisotropic (Ethayarajh, 2019; Mu and Viswanath, 2018), rare tokens degenerate (Gao et al., 2019), and whitening improves semantic similarity (Su et al., 2021); INLP and LEACE (Ravfogel et al., 2020; Belrose et al., 2023) also whiten, but to remove a concept from representations, whereas we whiten to find it in the weights.

Multilingual Knowledge and Unlearning Multilingual LMs encode language identity in a few dominant directions (Chang et al., 2022; Libovický et al., 2020) and share conceptual knowledge across languages (Wendler et al., 2024), yet unlearning in one language often fails to carry over (Choi et al., 2024; Lu and Koehn, 2025).

9 Conclusion

EMBER often misses concept features because concept tokens share a common direction with other rare tokens. CINDER sharpens the embedding matrix toward the concept, lets each feature find its width and verifies features both ways, and one multilingual fit covers eight languages at an eighth of the judge calls. Under an evaluation that stops rewarding destroyed tokens, CINDER reaches more concepts and, at moderate edit size, removes more, and embedding erasure blocks access to a concept through its words. Where the edited words must stay usable, we recommend CINDER's centred edit: CINDER-z scores higher only where the evaluation does not require the words to stay readable (§7).

Limitations

Model and language coverage Single-language results use one small model (Gemma-2-2B-it) in English, and multilingual results use one model (Llama-3.1-8B-Instruct) with a 4-bit quantized transformer and fp16 embeddings. Whether the findings transfer to other models remains to be tested.

Abstract concepts Concepts that lie between broader topics, such as Stoicism or Gothic architecture, yield no concept-specific embedding feature under EMBER or CINDER (§6); embedding-level erasure, with either method, does not reach them.

Embedding edits in isolation We evaluate embedding edits on their own, not ensembled with MLP-based methods as EMBER recommends, and we do not test robustness to relearning, EMBER's main strength. Whether relearning robustness also depends on the edit size ($r \approx 1$ vs. $r \approx 100$) is an open question.

Evaluation material Efficacy and collateral are multiple-choice accuracies, and collateral is lenient, since the options and context can rescue a damaged word. Collateral questions, neutral prompts and free-text labels were written and checked by LLM agents, whose blind spots may be correlated, and no human spot check was done; non-English questions are machine-translated.

Multilingual evaluation The multilingual study follows EMBER's protocol (its δ grid and metrics, with perplexity in place of Alpaca) and has no collateral term, so, like the standard protocol, it cannot tell erasing a concept from destroying its words. On Gemma's three development concepts, the mean term that CINDER-z and EMBER add to their edits erases mostly by damaging the prompt (§7), so part of multilingual CINDER-z's gain over CINDER, and of every approach's score, may be such damage; the mean-only test was not repeated on Llama.

Statistical power With 11–12 held-out concepts as the unit of analysis, differences below about 0.08 are hard to resolve, and several supporting analyses were defined after the primary result was known; we label them post hoc.

Sharpening target Contrastive Sharpening uses the concept's own Wikipedia page as its target, which overlaps the concept sentences. Targets disjoint from the sample keep the feature-level gains (§6), but the erasure experiments use the page target, and η and γ were chosen with it.

Ethical Considerations

CINDER is designed to remove hazardous, private or copyrighted knowledge more reliably. Like other editing techniques, it carries a dual-use risk: an effective erasure method could be used to suppress legitimate information. Our evaluation findings also carry a caution: embedding-level erasure, as commonly evaluated, can appear effective while only blocking access through specific words and damaging their other meanings, so it should not be relied on alone to remove dangerous knowledge.

AI Disclosure and Reflection

The research questions, hypotheses and main ideas of this work are our own; we used AI tools to support them, not to replace them. We used Claude Code (Anthropic) as a research assistant: to brainstorm and critique our ideas, to implement and debug the experiment code, and to scale the experiments to 38 concepts, several seeds and eight languages on the cluster. In the final experimental stages, once we were satisfied with their direction, we let it make routine decisions within the plan we had set, and reviewed them afterwards. It also helped draft and revise this report; we read every part of it, marked what to change, and set its style, its framing and the negative results it reports. LLMs also produced part of the evaluation material: Claude agents wrote and checked the collateral questions and neutral prompts and labelled features and free-text answers without seeing the method (§5.1), and Gemini 3.5 Flash-Lite generated and verified the questions of the 20 new multilingual concepts. Within the method, Gemini 3.5 Flash-Lite is the feature judge, the lexicon generator and the coherence judge (EMBER used Gemini 2.5 Flash-Lite). We take full responsibility for every claim in the paper.

Reflection The assistant was most useful for implementation and scale: it let us test each idea on many concepts and seeds rather than a few examples. The ideas behind CINDER, however, came from working with the data by hand: we wrote Jupyter notebooks with the assistant's help, ran them ourselves, inspected the features and the evaluations, and experimented. We found no substi-

tute for this, and our impression is that the assistant could not have proposed these directions on its own. It was least reliable where judgement was needed. It stated findings with more confidence than it had, so we asked it to give a confidence level with each claim; its drafts were dense; numbers drifted from the result files, and two audits of the text against them corrected about 40 claims; and it sometimes acted before we had agreed, starting an analysis we had only discussed or editing text we had asked only to discuss. We therefore kept the decisions with us and checked its claims against the code and the results.

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A EMBER: Reproduction Details

Sparse MF We follow EMBER's App. A.1. Given $A \in \mathbb{R}^{d\times n}$, alternating least squares updates $Y \leftarrow (Z^\top Z+\lambda I)^{-1}Z^\top A$ and $Z \leftarrow AY^\top(YY^\top+\lambda I)^{-1}$ with ridge $\lambda = 10^{-4}$. After each Y update the sparsity rule is applied to the dense Y (row sparsity: the top $\lceil s\,n\rceil$ entries of each row by magnitude, $s = 0.01$), and each row of Y is rescaled to unit norm with the scale absorbed into Z. Z and Y start from i.i.d. $\mathcal{N}(0,1)$ draws; we stop when the relative error improves by less than 10^{-4} for 200 iterations and the support changes in fewer than 0.5% of its entries, or after 20,000 iterations. $k = 100$ on Gemma-2-2B-it ($d = 2304$) and $k = 200$ on Llama-3.1-8B-Instruct ($d = 4096$). Embeddings are the input embedding rows scaled by $\sqrt{d}$ for Gemma, as the model uses them.

Data EMBER's 18 concepts: Ancient Rome, Artificial Intelligence, Baseball, Cannabis, COVID-19 pandemic, Culture of Greece, Gambling, Golf, Gun, Halloween, Harry Potter, Heroin, Nazism, Pornography, Republic of Ireland, Uranium, Valentine's Day, World War II. $\mathcal{S}_C$ holds 300 sentences sampled from the concept's Wikipedia page and $\mathcal{S}_N$ 300 sentences from random English Wikipedia pages (sampling seed 42); for Harry Potter $|\mathcal{V}^*| = 4{,}780$. Questions follow EMBER: 50 validation and 50 test multiple-choice questions per concept and per similar domain, and MMLU (Hendrycks et al., 2021). The 20 additional concepts are listed in App. F.

Feature selection The 20 features with the largest ρ (Eq. 1) and $\rho > 2$ go to the judge, which uses EMBER's two prompts (its Figures 4–5) on each feature's top 20 concept-labelled tokens and accepts at confidence ≥ 0.85. Our judge is Gemini 3.5 Flash-Lite; EMBER used Gemini 2.5 Flash-Lite. Re-judging 230 candidates flipped 0.4% of decisions.

Fidelity On EMBER's protocol our reproduction reaches concept, similar-domain and MMLU accuracy of 48.4, 83.2 and 55.1, against 52.1, 84.6 and 53.6 in the paper's Table 6. Tuning with Alpaca coherence inside the δ grid, as the paper does, gives the same δ and result as our cheaper MMLU gate on Harry Potter ($H = 0.727$ vs. 0.726). In an earlier phase we found that a 4-token answer budget in the original multiple-choice scorer counted many correct answers as wrong at large edit strengths, inflating EMBER's efficacy. All erasure numbers here use the repaired scorer, except the cross-lingual leakage of the first study (App. F), which we report with the original scorer and with unparseable answers dropped.

Why the protocol misses token destruction Over the 344 points of the first erasure run (EMBER and six intermediate variants on 14 concepts, 8 strengths each, seed 42), MMLU never falls below 0.93 of the unedited model, so the 0.9 gate never rejects a strength, and Alpaca coherence stays at or above 0.92 at every selected point; the similar-domain questions avoid the edited words. We wrote 20 neutral prompts per development concept, most using an edited word in another sense (Prince Harry, Caesar salad) and a few on adjacent topics with no edited word, and compared each greedy answer with the unedited model's. At $\delta = 200$, EMBER breaks 12, 3 and 7 of 20 on Harry Potter, Ancient Rome and Valentine's Day (seeds 42, 44 and 42), and CINDER 9, 12 and 9; at $\delta = 1$, both break 0–1 on each. $\delta = 200$ is EMBER's selected strength on Harry Potter; for CINDER, which selected $\delta = 2$ there, it is a stress test.

B Contrastive Sharpening: Details

B.1 Construction

Background The background is the set of token types in a sample of English Wikipedia, each type counted once (107,472 types for Gemma), so that a direction is loud in the background because many *different* tokens use it, not because a few frequent ones do. We compute its mean μ_B and covariance $\Sigma_B = Q\Lambda Q^\top$ once per model. The ridge $\varepsilon = 10^{-4}\cdot\mathrm{mean}(\Lambda)$ only damps the smallest eigenvalues.

Target The sharpening target is the set of token types of the concept's English Wikipedia page ($m = 3{,}124$ for Harry Potter), whitened with Eq. 3. $\eta = 0.05$ and $\gamma = 1$ were chosen by a judge-free sweep on Harry Potter, Ancient Rome and Valentine's Day: each of five values of η was ranked by

concept-vs-neutral AUC, concept-only purity of the top feature, concept-token coverage and $\rho_{\max}$, and the lowest mean rank won. $\eta = 1$ is whitening only; γ sets how strongly enriched directions are stretched.

Exact inverse In column form, $\mathbf{x}_t = T(\mathbf{e}_t - \mu_B)$ with $T = S_\eta^{\gamma/2} W$, which is invertible because $S_\eta \succeq \eta I$ and $W \succ 0$. If sparse MF gives $\mathbf{x}_t \approx \sum_i Y_{i,t} \tilde{\mathbf{z}}_i$, then $\mathbf{e}_t - \mu_B \approx \sum_i Y_{i,t} T^{-1} \tilde{\mathbf{z}}_i$: the same coefficients reconstruct the original embedding with the mapped-back dictionary $\mathbf{z}_i = T^{-1}\tilde{\mathbf{z}}_i = W^{-1} S_\eta^{-\gamma/2} \tilde{\mathbf{z}}_i$, and the mean cancels in any difference. Every erasure run checks numerically that the transformed embedding rows equal the factorized matrix columns before editing.

B.2 Why sharpening finds the concept

Let $M = \frac{1}{m}\sum_j \tilde{\mathbf{u}}_j \tilde{\mathbf{u}}_j^\top = P \operatorname{diag}(\lambda_j) P^\top$ be the whitened second moment of the target, so that $S_\eta = P \operatorname{diag}(g_j) P^\top$ with $g_j = (1-\eta)\lambda_j + \eta$.

Proposition 1. *(i) The eigenvalues of M are the generalized eigenvalues of the pair $(\Sigma_C + \boldsymbol{\delta}\boldsymbol{\delta}^\top, \Sigma_B + \varepsilon I)$, i.e. the stationary values of*

$$R(\mathbf{v}) = \frac{\mathbf{v}^\top \big(\Sigma_C + \boldsymbol{\delta}\boldsymbol{\delta}^\top\big)\mathbf{v}}{\mathbf{v}^\top (\Sigma_B + \varepsilon I)\,\mathbf{v}}, \qquad \boldsymbol{\delta} = \mu_C - \mu_B,$$

where μ_C and Σ_C are the mean and covariance of the page's embeddings, attained at $\mathbf{v}_j = W\mathbf{p}_j$. (ii) For any set of tokens with whitened second moment G, the second moment of the sharpened tokens along $\mathbf{p}_j$ is $g_j^\gamma\, \mathbf{p}_j^\top G\, \mathbf{p}_j$. In particular, background tokens, for which $G = I$ up to ε, carry the same energy along every direction after whitening and g_j^γ after sharpening, and the target's own energy along $\mathbf{p}_j$ becomes $g_j^\gamma \lambda_j \approx (1-\eta)^\gamma \lambda_j^{1+\gamma}$ for $\lambda_j \gg \eta$.

Proof sketch. (i) $\mathbf{p}^\top M \mathbf{p} = \mathbf{p}^\top W(\Sigma_C + \boldsymbol{\delta}\boldsymbol{\delta}^\top)W\mathbf{p}$, since the target's second moment around μ_B is $\Sigma_C + \boldsymbol{\delta}\boldsymbol{\delta}^\top$; with $\mathbf{v} = W\mathbf{p}$ and $\mathbf{p}^\top\mathbf{p} = \mathbf{v}^\top W^{-2}\mathbf{v} = \mathbf{v}^\top(\Sigma_B + \varepsilon I)\mathbf{v}$, the Rayleigh quotient of M equals $R(\mathbf{v})$. (ii) $S_\eta^{\gamma/2}\mathbf{p}_j = g_j^{\gamma/2}\mathbf{p}_j$, so $\mathbf{p}_j^\top S_\eta^{\gamma/2} G S_\eta^{\gamma/2}\mathbf{p}_j = g_j^\gamma\, \mathbf{p}_j^\top G \mathbf{p}_j$. □

Since the error that a sparse-MF feature removes is the energy of its support along it, the proposition says which directions sparse MF will find profitable. In the raw matrix they include the mean direction that rare tokens share, so a concept's rare tokens are grouped with other rare tokens. After whitening, all background directions are equally profitable, so the concept's tokens form a pure cluster of ordinary value; whitening alone finds a concept feature less often than the raw matrix (on Harry Potter it merges the concept with magic). After sharpening, they are ordered by the concept-to-background ratio, raised to the power $1 + \gamma$ for the concept's own tokens. Two further facts make this precise. First, EMBER's alternating fit depends on A only through the token Gram matrix $K = A^\top A$: with $G_Y = YY^\top + \lambda I$, the dictionary update gives $Z^\top Z = G_Y^{-1} Y K Y^\top G_Y^{-1}$ and $Z^\top A = G_Y^{-1} Y K$, so the dense solve, the masks and the error are functions of K (verified numerically: identical supports for A and QA, Q orthogonal). Under CS with $\gamma = 1$, K becomes a *page kernel*, $K(s,t) = (1-\eta)\frac{1}{m}\sum_{j\in\text{page}} k(s,j)\,k(t,j) + \eta\, k(s,t)$, with k the background-Mahalanobis inner product: two tokens are similar when they resemble the same tokens of the concept page. Second, with the other features fixed, the best feature on a support S of fixed size lowers the error by $\sigma_1(R_S)^2 = \big(\sum_{t\in S}\|\mathbf{r}_t\|^2\big)\,\kappa(S)$, the support's energy times its coherence $\kappa(S) \in [1/|S|, 1]$. A concept can therefore fail in two ways: it is *impure* if other tokens project more strongly along its best direction (the raw case, where rare tokens share $\hat{\mu}_B$), or *unprofitable* if its energy times coherence is below what other directions offer (the whitened-only case). Across concepts the largest gain is $g_1 = 15.7$ (median; range 10.9–29.3), and 870 of 2,304 directions have $g_j > 1$.

Decomposition Medians over 38 concepts × 5 seeds, row sparsity:

	raw	$\perp \hat{\mu}_B$	centred	whit.	CS
Concept found (% runs)	56	61	59	49	69
Its neutral tokens	4.5	4.5	4.5	4.0	3.0
Its $\|\mathbf{z}\|^2$ percentile	86	66	41	54	58
Seeded: purity	.24	.27	.04	.86	.86
Seeded: value pct.	100	97	100	70	91
SVD rank matching	4	22	4	66	49

The seeded feature starts on the concept's lexicon tokens and alternates between the top singular vector of its support and the top tokens along it until the support repeats.

B.3 Is the gain circular?

Design 18 EMBER concepts, seeds 42–44, row sparsity (global sparsity as a secondary analysis), everything fixed except the sharpening target, preregistered before any new fit. Targets: the page

(default); *disjoint*, page sentences that are not near-duplicates of any sampled sentence; *related*, four other Wikipedia pages about the concept, taken by rule from the page's "Main", "Further" and "See also" links and size-matched; a random page subset as large as *disjoint*; the leak-free control (page minus every token of $\mathcal{V}^*$); and *wrong*, the page of an unrelated concept. A token is *strict* if it is concept-only and either a lexicon word or occurs at least twice in $\mathcal{S}_C$ and is at least ten times more frequent there than in Wikipedia.

Results Strict purity of the best concept feature (raw .568): page $+.052$ ($p = .088$), disjoint $+.060$ (.003; Holm .034), related $+.062$ (.009), page subset $+.046$, leak-free $-.064$ (n.s.), wrong $-.445$ ($p < 10^{-4}$). Separability of strict and neutral tokens (AUC, raw .576): page $+.055$ (17 of 18 concepts), related $+.041$ (17 of 18), disjoint $+.037$ (15 of 18); on tokens absent from the target, related still gives $+.034$ ($p = .0003$). With the wrong page the concept feature is found in 2 of 54 runs (raw 39). The coverage measure used to choose η rises only with the page target ($+.033$) and falls with the disjoint and related ones, so it was the circular quantity. No target but the wrong one changes the ρ-rank of the concept feature significantly (mean 4.9 raw, 3.9 page), although sharpening lifts concepts that the raw fit buries (Ancient Rome 22 $\rightarrow$ 2, Harry Potter 14 $\rightarrow$ 1, Halloween 15 $\rightarrow$ 1.3). The target also steers which side of the concept is amplified: for Harry Potter, the page keeps the proper-name feature, while the disjoint and related targets yield a "magic" feature, both ranked first. Under global sparsity the purity gain disappears for every target, page included, while the separability gains remain.

Sensitivity to η and γ On the 15 concepts outside the tuning set (page target, 3 seeds), every setting of $\eta \in \{0.05, 0.2, 0.5\}$ and $\gamma \in \{0.5, 1, 2\}$ beats the raw matrix on strict purity, separability and ρ-rank; the default ranks 6th of 9, $\gamma = 2$ is slightly better ($+.024$ purity), and $\gamma = 0.5$ is worst on every measure.

B.4 Models with a larger embedding dimension

The question S_η is estimated from the m token types of one page. On Llama-3.1-8B ($d = 4096$), 15 of the 18 English pages have $m < d$ (median $m/d = 0.77$; Gemma 1.29), so $(1 - \eta)M$ has a null space that sharpening shrinks to $\eta^{\gamma/2}$.

Rank is not the problem The fitted features put under 0.5% of their mapped-back norm into these floor directions; the three Llama pages with $m > d$ fail like the others; and Gemma pages cut below d, or Gemma's own pages with $m < d$, work as before.

Noise is Only about 12 eigenvalues of Llama's whitened page moment exceed the largest eigenvalue of the same moment computed from m random background types (Gemma: 74), and in the noise-dominated bulk each page token's own outer product supplies about half of its sharpened energy (predicted $d/(d+m)$; measured median 0.54, correlation 0.99 across 36 model–concept pairs). Since almost every concept-only token of $\mathcal{V}^*$ is on the page, sharpening then amplifies page membership rather than the concept: on Llama, 92% of features pass $\rho \geq 2$ (EMBER 32%), their words are close to incoherent, and the concept feature falls from a ρ-rank of about 10 to about 38 ($p = 10^{-4}$, 18 concepts $\times$ 3 seeds). No $\eta \in [0.05, 0.5]$ or $\gamma \in \{0.5, 1\}$ changes this, since the echo sits in the bulk, not on the floor.

Rule Sharpen only the resolvable directions, $S = I + \sum_{j:\lambda_j > \hat{\lambda}_{\text{edge}}} \left((1-\eta)\lambda_j + \eta - 1\right) \mathbf{p}_j \mathbf{p}_j^\top$, with $\hat{\lambda}_{\text{edge}}$ the largest eigenvalue of the random-type moment (the Marchenko–Pastur edge $(1 + \sqrt{d/m})^2$ is a close closed form). On Llama this gives a concept-feature rank of about 5 against 10 for EMBER ($p = 0.015$) with EMBER's coherence, though no better than whitening alone; on Gemma it keeps 47–98 directions and changes nothing. Pooling the concept's text over eight languages raises the number of resolvable directions to about 85, which, with the same rule, also works (rank 4.5). The multilingual fits of §4.4 were run before this analysis and use the full S_η on the pooled target, without the rule; in this judge-free test, the pooled target without truncation still lets 82% of features pass $\rho \geq 2$ (concept rank 15.5).

C Global Sparsity: Details

Algorithm Algorithm 1 replaces EMBER's per-row top-k step; everything else is unchanged. The budget equals EMBER's total (1% of all entries); the floor is 0.2% and the cap 3% of the tokens per feature ($m_{\min} = 10$, $m_{\max} = 144$ on Harry Potter). The cap exists for generic features; no concept feature reached it in 190 runs over 38 concepts (widest: Culture of Greece, 120 of 156 to-

Algorithm 1 Sparse MF with Global Sparsity (one iteration)

Require: $A \in \mathbb{R}^{d\times n}$, $Z \in \mathbb{R}^{d\times k}$, budget $K = \lceil s\,n \rceil k$, floor $m_{\min}$, cap $m_{\max}$

$Y^{\mathrm{d}} \leftarrow (Z^\top Z + \lambda I)^{-1} Z^\top A$ ▷ dense update

$s_{ij} \leftarrow |Y^{\mathrm{d}}_{ij}| \,/\, \|Y^{\mathrm{d}}_{i,:}\|_2$ ▷ scale-free share

$\mathcal{S} \leftarrow$ the top $m_{\min}$ entries of every row by s_{ij}

add the remaining entries in decreasing s_{ij}, skipping rows with $m_{\max}$ entries, until $|\mathcal{S}| = K$

$Y \leftarrow Y^{\mathrm{d}} \odot \mathbb{1}_{\mathcal{S}}$; rescale rows of Y to unit norm, absorb scales in Z

$Z \leftarrow AY^\top (YY^\top + \lambda I)^{-1}$

kens).

Variants All five rules below share the matrix, the budget and the fitting loop.

Rule	Support	Role
RS-1%	top 1% of each row	EMBER
RS-2%	top 2% of each row	more tokens, no reallocation
GS-norm	global top-K of $\lvert y_{ij}\rvert/\lVert \mathbf{y}_i\rVert$	CINDER
GS-gain	global top-K of $\lvert y_{ij}\rvert\,\lVert \mathbf{z}_i\rVert$	favours loud features
GS-entropy	width $\propto$ effective support $e^{H(\mathbf{y}_i)}$	ablation

An ℓ_1-penalized variant (the Lasso, Tibshirani 1996), tuned to the same number of nonzeros, never produced a clean Ancient Rome feature in 5 seeds and was dropped.

Results On the sharpened matrix, 18 concepts × 5 seeds (cap 5% in this sweep):

	RS-1%	RS-2%	GS-norm	GS-gain	GS-ent.
Width	43–52	86–104	15–113	9–143	25–121
Seed MAD	0	0	4.1	8.1	7.9
Clean runs (/90)	75	31	76	55	61
Neutral share	5.8%	8.2%	5.6%	8.0%	8.2%
Seed Jaccard	.719	.726	.735	.717	.733

MAD: mean absolute deviation of a concept's width across seeds. A run is clean when its concept feature has at most five neutral tokens. Only GS-norm adapts the width per concept while staying as clean and stable as EMBER's rule. This sweep used a 5% cap; with the current 3% cap, concept widths range over 21–114 tokens (concept means).

C.1 Widths and the Natural Tail

We use the saved fits of 38 concepts × 5 seeds (EMBER; CS with row sparsity; CS+GS; raw GS; CS with 2% row sparsity), 950 fits in all. *Dense re-solve.* With the saved dictionary fixed, $Y' = (Z^\top Z + \lambda I)^{-1} Z^\top A$ equals Y on the support in every fit (maximum error $2.2 \cdot 10^{-6}$), and every GS support is exactly the set of tokens whose share exceeds a single threshold per fit. *Natural tail.* Along the dense ranking of the concept feature (identified by lexicon hits), a token is *page-enriched* if it occurs in $\mathcal{S}_C$ and is at least 20 times more frequent there than in Wikipedia; the natural tail T is the first rank at which fewer than half of the next 20 tokens are page-enriched, fixed before any width statistic was computed. *Results.* On the sharpened matrix the median T is 69 tokens against a row-sparse width of 47; the 1% cut is short for 25, long for 8 and about right for 5 of 38 concepts (raw matrix: 15, 15, 8). Doubling the budget moves T by $+3$ tokens (rank correlation 0.96). GS widens 26 of the 28 features cut short; its width correlates with T at 0.67 [0.42, 0.84] against 0.27 for the row-sparse width, with mean error 30 vs. 42 tokens ($p = 0.0009$). This holds for all 45 non-degenerate definitions of the tail in a grid of 72, but GS compresses the range (slope 0.30): it overshoots narrow concepts by 24 tokens and undershoots broad ones by 44 (e.g., Buddhism 100 vs. 182). *Stability.* The width's seed deviation is 4 tokens when the same feature is identified (7.7 overall), and the width is unrelated to the feature's index ($\rho = -0.004$). *Breadth.* GS width does not correlate with counts of concept tokens (0.05); that count ranks concepts exactly like $|\mathcal{V}^*|$, since $\mathcal{S}_N$ is shared. *Direction.* The GS and row-sparse concept features have cosine 0.97 (within seed noise); raw and sharpened features, mapped back, only 0.75. *GS needs CS.* On the raw matrix, the width that GS adds beyond the row-sparse width is 16% neutral (against 7% for its first 47 tokens), and raw GS does not track T better than row sparsity ($p = 0.39$). Under GS the concept feature ranks first by ρ less often than under CS with row sparsity (18% vs. 38% of runs), although its wider support holds more concept tokens (54 vs. 41.5).

D Two-Way Judge: Details

Lexicon Once per concept, before any feature is judged, the LLM receives:

> *Given the concept '{concept}', list the {n} words most strongly associated with it. Rules: Start with the concept's own name and its basic word forms when they are single words [...]. Then the entities, people, places, objects, actions and terms that an ordinary person with general knowledge would immediately connect to this concept. Prefer well-known words over expert or niche terminology. Every word must be specific*

to this concept: avoid words that fit its broader category just as well (e.g. for 'World War II' not 'war', 'battle' or 'soldier'). Single words, lowercase, no duplicates or mere plural variants. Return ONLY a JSON list of {n} strings.

with $n = 20$. The prompt's examples name four concepts (Culture of Greece, Ancient Rome, Artificial Intelligence, World War II), which we exclude when evaluating the judge. In the multilingual pipeline the same call returns 20 words in each of the eight languages.

Feature words For each of a feature's top tokens we take the word in $\mathcal{S}_C$ that most often produces it (*ffindor* $\rightarrow$ *gryffindor*), lowercase and deduplicate the words, and keep the first 20; EJ@t accepts when at least t of them are lexicon words.

Variants

Rule	Accepts when
TJ	EMBER's two-stage judge accepts
WJ	TJ, shown each token's source words
WJ+share	WJ, and the concept share is ≥ 0.5
EJ@t	$\geq t$ lexicon words among the feature's words
TWJ	TJ $\vee$ EJ@5
TWJ-R	WJ+share $\vee$ EJ@4

Labelled evaluation Every rule scored every candidate feature of the CS+GS fits (23 concepts $\times$ 5 seeds, 7,105 candidates). Every feature accepted by some rule was labelled, without seeing the rules' decisions, as P (specific to the concept), B (broader or adjacent, e.g. nuclear physics for Uranium) or N (unrelated). Excluding the four prompt-example concepts leaves 19 concepts and 95 runs; a run is hit when an accepted feature is positive.

	P only			P + B		
Rule	Prec.	Rec.	Hit/72	Prec.	Rec.	Hit/91
TJ	.85	.69	60	1.00	.41	61
WJ	.56	.81	61	.77	.55	66
EJ@5	.74	.64	64	.99	.43	81
TWJ	.71	.77	67	.99	.54	82
TWJ-R	.61	.90	71	.91	.68	84

Recall counts only positives that some rule accepted. The threshold $t = 5$ was chosen on this set. The 15 concepts added after the prompt was fixed were picked informally rather than from earlier erasure benchmarks, and they are harder for every judge: TJ finds the concept or a broader version of it in 34 of 75 runs, against 61 of 91 on the first 19 concepts. On them, EJ@5 does not generalize (precision 0.46, recall 0.26) and TWJ falls to 0.53/0.56, whereas TJ $\vee$ WJ+share keeps precision 0.71 and recall 0.89 (49 of 54 runs hit, against 33 for TJ). In erasure the two judges perform the same. Table 4 gives the comparison on the held-out concepts (75 runs, 54 of them with a P candidate, 70 P-labelled candidates).

	P only			P + B		
Rule	Prec.	Rec.	Hit/54	Prec.	Rec.	Hit/75
TJ	.67	.53	33	.98	.21	34
WJ+share	.79	.87	48	.99	.30	51
EJ@5	.46	.26	15	1.00	.15	28
TWJ	.53	.56	35	.99	.29	40
TJ $\vee$ WJ+share	.71	.89	49	.98	.33	52

Table 4: Judges on the 15 held-out concepts (CS+GS fits, 5 seeds, top-20 candidates). Hit: runs with an accepted positive feature, of the runs that have one.

TJ's wrong rejections on the held-out concepts TJ rejects 33 of the 70 P-labelled candidates, and TJ $\vee$ WJ+share recovers 25 of them. The rejections concentrate on five concepts (Table 5), and the stage-1 descriptions show why: the description names the feature's spelling, script or the broader sense of its words rather than the concept, and stage 2 then rejects it. TJ rejects no P candidate of Doctor of Philosophy, Robot, Surfing or Tourism, and American Revolutionary War, Louvre and Prime number have none.

Cost TJ makes two LLM calls per candidate feature, one that describes its top tokens and one that decides from the description. In the single-language study it judges up to 20 candidates per run (the 20 features with the largest ρ above 2), so up to 40 calls; in the multilingual study it judges the top 10 of each fit, so 20 calls per fit, which gives the 20 and 160 calls per concept of Table 2. The lexicon path makes one call per concept, shared by all seeds and fits, and its matching is deterministic, so its decisions are reproducible and free to recompute after any change to the factorization.

E Centred and Uncentred Edits

Read strength The first thing the model does with an embedding is normalize it and project it into queries, keys and values. For a unit direction $\mathbf{u}$ we define its read strength as $\left\|[W_Q; W_K; W_V]\,(\mathbf{g} \odot \mathbf{u})\right\|$, with the layer-0 attention projections stacked and $\mathbf{g}$ the layer-0 RM-

Concept	Rej.	Runs	TJ's stage-1 description (stage-2 confidence)
Karate	7/7	$0 \to 5$	tokens starting with the letter 'k' (0.99–1.0); letters 'b' and 'u' (0.99–1.0); features of *kendo, kihon, shotokan, kyokushin, karateka*
Flush toilet	9/10	$1 \to 5$	sanitation, waste elimination and restrooms (0.75–0.85); sub-words starting with 'fl' (0.99–1.0) for *flush, flushing, flushometer*
Existentialism	5/6	$1 \to 4$	being, existence and living (0.85–0.95) for *existence, existentialism, existenzphilosophie*
Sigmund Freud	5/5	$0 \to 1$	analysis, examination and interpretation (0.95) for *psychoanalysis, psychoanalyst, analytic*
Judaism	4/10	$5 \to 5$	Hebrew vowels, diacritics and letters (0.85–0.95) for Hebrew-script religious terms
Cloud	1/5	$4 \to 5$	atmospheric phenomena such as clouds, mist and smoke (0.85)
Giraffe	1/2	$1 \to 2$	African savanna wildlife, long-necked animals (0.85)
The Matrix	1/5	$4 \to 5$	"the science fiction film franchise *The Matrix* and its associated digital or numerical grids" (0.85)

Table 5: TJ's wrong rejections on the held-out concepts (our recount of the saved labels). Rej.: P-labelled candidates TJ rejects, of all P candidates; Runs: runs hit by TJ $\to$ by TJ $\vee$ WJ+share (of 5). Descriptions are paraphrased from the stored stage-1 answers, except the quoted one.

SNorm scale, relative to the mean over 500 random unit directions; for Llama we also stack the MLP's gate and up projections, as the multilingual study did. For a method we take the median over its edited tokens' directions $\mathbf{c}_t/\|\mathbf{c}_t\|$.

The dictionary step does not depend on the metric Write $A_c = A - \mu_B \mathbf{1}^\top$ for the centred embeddings. The sharpened fit alternates $\tilde{Z} \leftarrow \tilde{X}Y^\top(YY^\top + \lambda I)^{-1}$ on $\tilde{X} = TA_c$ with a masked coefficient update, and CINDER maps the dictionary back as $T^{-1}\tilde{Z}$.

Proposition 2. *For invertible T, $B \in \mathbb{R}^{d\times n}$, $Y \in \mathbb{R}^{k\times n}$ and $\lambda \geq 0$ with $YY^\top + \lambda I \succ 0$,*

$$\arg\min_Z \|T(B - ZY)\|_F^2 + \lambda\|TZ\|_F^2 = BY^\top(YY^\top + \lambda I)^{-1},$$

independently of T.

Proof. $Z \mapsto \tilde{Z} = TZ$ is a bijection, and in $\tilde{Z}$ the objective $\|TB - \tilde{Z}Y\|_F^2 + \lambda\|\tilde{Z}\|_F^2$ is strictly convex with minimizer $\tilde{Z}^* = TBY^\top(YY^\top + \lambda I)^{-1}$; hence $Z^* = T^{-1}\tilde{Z}^* = BY^\top(YY^\top + \lambda I)^{-1}$. □

At convergence of the fit, the mapped-back dictionary is therefore the Euclidean least-squares dictionary of the *centred* embeddings, $Z_c = A_cY^\top(YY^\top + \lambda I)^{-1}$, for the affine model $\mathbf{e}_t \approx \mu_B + Z\mathbf{y}_t$ (on Gemma, $\|T^{-1}\tilde{Z} - Z_c\|/\|Z_c\| \leq 0.008$ in all 137 runs). The coefficient step is where the metric matters: Y is an M-orthogonal projection with $M = T^\top T$, so whitening chooses which tokens are grouped and how they are weighted.

Proposition 3. *CINDER-z's dictionary (Eq. 4) is $Z_z = Z_c + \mu_B\mathbf{b}^\top$ with $\mathbf{b} = (YY^\top + \lambda I)^{-1}Y\mathbf{1}$, so its edit is $\mathbf{c}_t^z = \mathbf{c}_t + \beta_t\mu_B$ with $\beta_t = \sum_{i\in\mathcal{F}_C} b_iY_{i,t}$.*

Proof. Substitute $A = A_c + \mu_B\mathbf{1}^\top$ in Eq. 4. □

In the data, $\mathbf{c}_t^z - \mathbf{c}_t$ has at least 0.9995 of its energy along μ_B in every run (Gemma 137, Llama 21). For a feature of width w with same-sign coefficients of size about $1/\sqrt{w}$, $\beta_t \approx 1$ whatever w, so at $\delta = 1$ an uncentred edit subtracts roughly one full μ_B (observed median β: EMBER 0.66, CINDER-z 0.56, CINDER 0.05; the edited tokens' own coordinate along μ_B is 1.06). The affine model also reconstructs raw embeddings better than the uncentred one (relative error 0.917 vs. 0.932, 137 of 137 runs).

Where the read-strength gap comes from Concept means on Gemma-2-2B; $\perp \mu$: with $\hat{\mu}_B$ projected out of the edit.

Edit	Read	$\hat{\mu}_B$ share	Read $\perp \mu$	$\cos(\mathbf{c}_t, \mathbf{e}_t)$
EMBER	1.86	0.44	1.31	0.50
EMBER, centred Z	1.09	0.07	1.31	0.35
CINDER	1.18	0.05	1.32	0.30
CINDER-z	1.99	0.48	1.32	0.50

EMBER vs. CINDER: $+0.68$ read (27 of 27 concepts), but -0.01 once $\hat{\mu}_B$ is removed ($p = 0.44$), with the same Rayleigh ratio of the background covariance along the edit (4.3 vs. 4.4). Across all edits, read strength follows the share along $\hat{\mu}_B$ (Spearman 0.75) more closely than the background variance along the edit (0.63). The mean term is also most of what aligns an uncentred edit with its token: with $\hat{\mu}_B$ removed from both, $\cos(\mathbf{c}_t, \mathbf{e}_t)$ is 0.31 for EMBER and 0.29 for CINDER ($p = 0.15$). The read strength of the

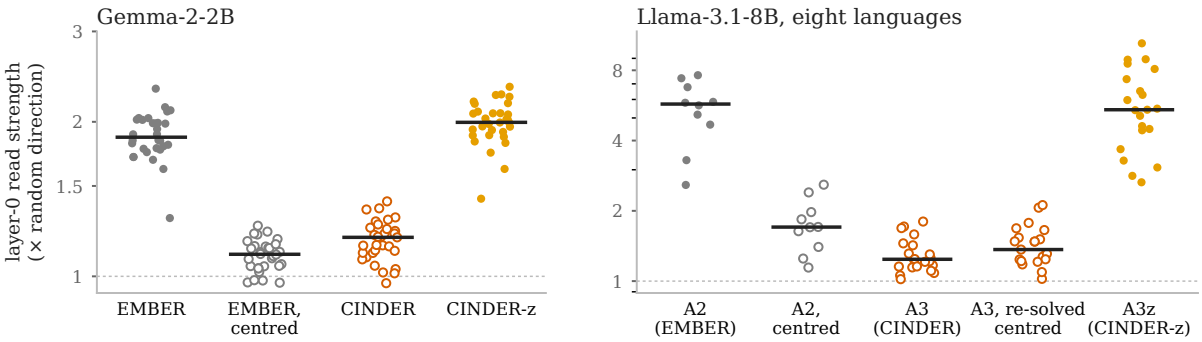


Figure 4: First-layer read strength of each edit relative to a random direction, per concept (dots) with medians (bars), on Gemma-2-2B (left) and the multilingual Llama-3.1-8B fits (right). Centring EMBER's dictionary brings its read strength down to CINDER's on Gemma and most of the way on Llama; adding the mean term back (CINDER-z, A3z) raises it again. Filled: uncentred edits; open: centred.

eigendirections of Σ_B is U-shaped in their variance (top decile 1.08, middle deciles 0.81–0.88, bottom decile 1.59, lowest 1% 3.4); strongly read directions concentrate on a few coordinates with a large first-layer normalization weight. On Llama, A3z exceeds A3 by $+4.48$ in read strength, of which $+4.34$ is the mean term; $\hat{\mu}_B$ is read at $20.4\times$, so a 7% share suffices.

What the first layer does with the mean Setting the layer-0 normalization weight to one lowers $\hat{\mu}_B$'s read gain from $2.93\times$ to $2.10\times$ and the EMBER–CINDER gap by 53%; zeroing its eight largest entries (3.98–5.97, against a median of 1.17) lowers them to $1.74\times$ and 75%; with $\hat{\mu}_B$ projected out, every edit is read at 1.28–$1.35\times$ under all three settings. In the forward pass (50 tokens from 5 runs, each direction relative to random ones), the linear proxy predicts the measured layer-0 write well (Spearman 0.76–0.81), and the mean and uncentred directions push far more change into the *later* positions of the prompt (2.1–$4.3\times$, against 0.8–$1.2\times$ for the centred ones). Yet at small perturbations the centred directions change the token's own next-token distribution more (KL $5.4\times$ for CINDER, $3.9\times$ for centred EMBER, $1.6\times$ for the mean direction); at $s = 1$ the directions are close. The mean direction is thus "read strongly" in that it spreads into the context, not in that it changes what the model predicts at the token.

E.1 The Mean Term on Its Own

Design By Proposition 3, CINDER-z's edit is CINDER's plus $\beta_t \mu_B$, a term that carries no information about the concept. We therefore evaluated that term alone (MEAN-C: $\mathbf{e}'_t = \mathbf{e}_t - \delta\beta_t^z \mu_B$ on CINDER's tokens, with CINDER-z's own β^z), and the same split for EMBER (EMBER-c: EMBER's Y and $\mathcal{F}_C$ with the dictionary resolved on centred embeddings; MEAN-E: $\beta_t \mu_B$ with EMBER's coefficients), on the 8 development runs (Harry Potter, Ancient Rome and Valentine's Day; 7 with a CINDER edit, 6 with an EMBER edit), over the same grid of r and with the evaluation of App. G. The decompositions hold to a relative error of 10^{-4} in every run. The test was pre-registered before any result (`results/paper_analysis/mean_edit/PREREG.md`); with three concepts, its statistics are descriptive.

The mean term erases by breaking the prompt Over $r \geq 1$, CINDER-z removes $+0.155$ more than CINDER (7 of 7 runs), and the mean term alone removes more than CINDER-z at every $r \geq 2$ (validation efficacy 0.60–0.76 vs. 0.46–0.55) and more than a random direction of the same size ($+0.48$ at $r \geq 10$, 7 of 7; Figure 5). It does so by damaging the context: an unrelated multiple-choice question placed after a sentence with the edited word drops from 0.54 to 0.15–0.18 accuracy at $r \geq 10$, below chance, and collateral falls to 0.07. Across variants and sizes, efficacy above the deletion floor tracks this context damage (Spearman 0.54, 63 points). The centred edits destroy the words as completely at large r (copy 0), yet stay near the floor with the context intact. EMBER shows the same pattern: centring its dictionary removes its large-r gain ($+0.097$ over $r \geq 1$, 6 of 6) and its context damage, and raises efficacy at small edits ($+0.07$ and $+0.06$ at $r = 0.25$ and 0.5).

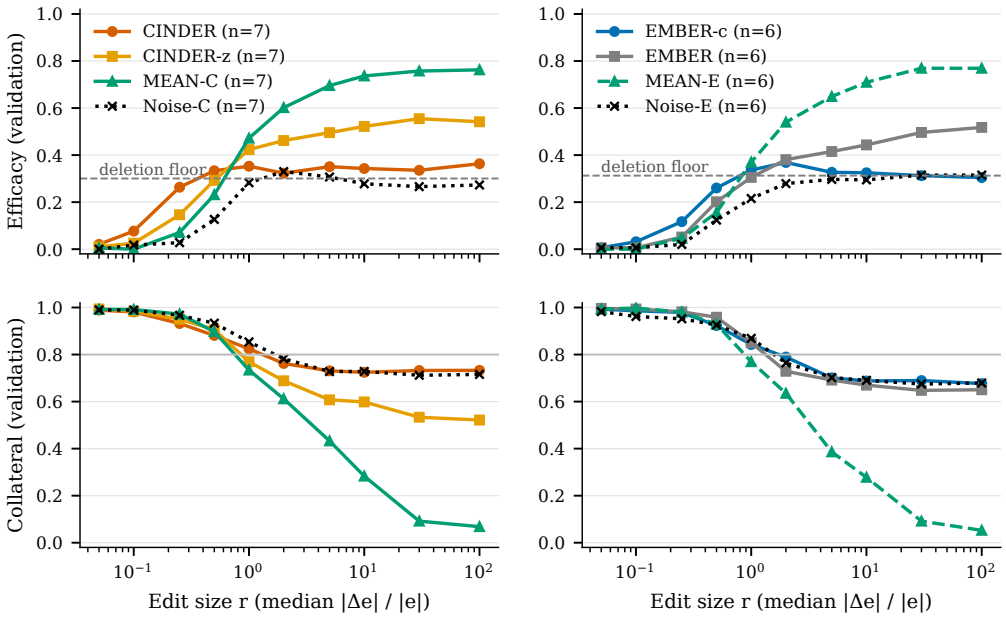


Figure 5: The mean term on its own (development runs, validation). Efficacy (top) and collateral (bottom) against edit size, mean over runs. Left: CINDER, CINDER-z, the mean-only edit MEAN-C $= \beta_t^z \mu_B$ on CINDER's tokens, and a random direction (7 runs). Right: EMBER, EMBER re-solved on centred embeddings (EMBER-c), the mean-only edit MEAN-E on EMBER's tokens, and a random direction (6 runs). Dashed: deletion floor of the token set; grey: the collateral gate.

Per unit of word damage, the centred edit erases most At the size where 90% of the edited words remain copyable, CINDER reaches efficacy 0.30, CINDER-z 0.20 and the mean term 0.13 (EMBER-c 0.22, EMBER 0.15, MEAN-E 0.11); at 50% the three CINDER variants are equal. Under the copy-gated rule, the mean term erases almost nothing (test efficacy 0.01, against 0.12 for CINDER and 0.10 for CINDER-z). Under the collateral rule, however, it ties both (test H_3 0.43 vs. 0.42 and 0.41): collateral questions do not register damage to the rest of the prompt at $r \approx 0.5$.

Why the direction effect vanishes at $r = 1$ The centred edits (CINDER, EMBER-c, and CINDER's direction on EMBER's tokens) reach a plateau near the deletion floor by $r \approx 0.5$–1, while EMBER is still climbing, and keeps climbing past the plateau through generic damage. On the 5 runs with all cells, centring EMBER's dictionary closes about a quarter of the gap to CINDER's direction at $r = 0.5$ and nearly all of it at $r = 1$; the mean term becomes a net gain for EMBER only at $r \geq 2$–5.

F Multilingual Erasure: Details

English erasure leaks In a first study we erased EMBER's 18 concepts from Gemma-2-2B-it with EMBER, SNMF (Shafran et al., 2026) and their combination, tuned every edit in English, and asked the held-out questions in English, German, Spanish, French and Russian. Retention, the share of the concept knowledge left (1 = as before, 0 = chance), was on average 0.24 higher in the four other languages than in English under EMBER (95% CI [0.15, 0.32]; 0.19 [0.12, 0.27] with unparseable answers dropped), and 0.41 (0.31) higher in Russian, where the English edit touches none of the concept's tokens. These numbers use the original scorer of App. A. SNMF, which edits MLP weights, erased all five languages about equally (+0.01), and adding EMBER to it brought the gap back (+0.16).

Languages, model and data Llama-3.1-8B-Instruct in its eight supported languages: English, German, French, Italian, Portuguese, Hindi, Spanish and Thai. The transformer runs in 4-bit NF4 and the edited embeddings in fp16, identically for every approach. Concept sentences come from the concept's native Wikipedia page (via interlanguage links), up to 300 per language, topped up with NLLB translations of English sentences for short pages; an equal number of neutral sentences comes from native Wikipedia pages. Shared fits factorize the tokens of 30 concept and 30 neu-

tral sentences per language (about 6,500 tokens), the fewest that give a full-rank matrix; with the dictionary fixed, a ridge least-squares solve followed by the fit's own support rule gives the coefficients of every token in the full sets, and CINDER-z then re-solves the dictionary on the raw embeddings with these coefficients fixed. The 20 held-out concepts are French Revolution, Tuberculosis, Cricket, Buddhism, Plutonium, Judaism, Karate, Existentialism, Prime number, The Matrix, Louvre, American Revolutionary War, Robot, Sigmund Freud, Giraffe, Surfing, Flush toilet, Cloud, Doctor of Philosophy and Tourism. EMBER's English questions are used as published; German, French and Spanish come from existing translations; Italian, Portuguese, Hindi and Thai are machine-translated (NLLB and MADLAD candidates ranked by LaBSE, with script and number checks). Questions for the new concepts were generated from the English page in EMBER's format, checked by a blind verifier and by automatic checks, and translated the same way.

Metrics and gates Per language L: efficacy (chance-corrected concept accuracy driven to chance), specificity (harmonic mean of Global-MMLU and similar-domain retention), fluency ($\min(1, \text{ppl}_{\text{orig}}/\text{ppl}_{\text{edit}})$ on 100 held-out native sentences), and H_L, their harmonic mean. A concept's score is the mean H_L over eligible languages (unedited accuracy $\geq$ 0.4). $\delta \in \{0.5, \ldots, 200\}$ is chosen on validation, one value for all languages, and fails if MMLU or fluency drops below 0.9 of the unedited model in any language. We report the *strict* score (0 if a gate fails on test) and the *ungated* one.

Approaches

	Fit	Selection	Edit
A1	8 per-language EMBER fits	TJ per language	each language's; overlaps resolved by page length
A2	1 fit on the 8-language sample, Y extrapolated	TJ	EMBER's
A3	A2 with CS (8-language background) and GS	TWJ	mapped back
A3z	as A3	as A3	Z re-solved on raw E

Results Table 6 gives the development run on EMBER's 18 concepts, on which A3z was defined (post hoc, after A3's result). The hypotheses were then pre-registered and tested on the 20 new concepts with the protocol frozen (Table 7): H1, that a single multilingual EMBER fit (A2) comes within 0.05 of per-language EMBER (A1) on the concepts both edit; H2, that the original-space edit (A3z) beats the mapped-back one (A3) on most of the concepts A3 edits, with a higher mean; and H3, that A3z is at least as good as A2 on the concepts all four approaches edit. On the 20 new concepts, only H2 can be tested with more than two concepts, and its gain has the same sign as in development ($+0.19$, 7–3, $p = 0.064$; ungated $+0.26$, 8–2, $p = 0.014$). Over all 38 concepts, H2 holds ($+0.166$, 95% CI $[+0.054, +0.268]$, 16–5, $p = 0.007$); H1 is consistent with the data but rests on 10 concepts (-0.051 $[-0.156, +0.057]$, 3–7, $p = 0.38$); and H3 is not supported (-0.005 $[-0.22, +0.17]$, 5–4, $p = 0.73$). Over the concepts that either approach edits (no edit scored 0), A3z beats A2 ($+0.213$ $[+0.053, +0.366]$, 15–4, $p = 0.011$) and ties A1 (-0.023 $[-0.156, +0.110]$, 15–13, $p = 0.78$).

Concept	A1	A2	A3	A3z
Cannabis	.41	–	.57	.85
Harry Potter	.79	.65	.26	.34
COVID-19 pandemic	.70	.42	.37	.73
Pornography	.64	.74	.28	0*
Baseball	.54	.52	.35	.69
Uranium	.58	–	.12	.15
Gun	.22	.39	.18	.53
Culture of Greece	.30	.27	.23	.55
Gambling	.54	.50	.22	.38
Ancient Rome	.42	–	–	–
Nazism	–	–	.09	.33
Golf	.33	–	–	–
Republic of Ireland	.24	–	–	–
Artificial intelligence	.22	0*	.33	0*
World War II	.27	–	–	–
Halloween, Heroin, Valentine's Day	–	–	–	–
Concepts edited (of 18)	14	8	11	11
Mean, strict	.344	.194	.166	.253
Mean, ungated	.344	.222	.166	.326

Table 6: Development run of the multilingual study: test score per concept on Llama-3.1-8B-Instruct (8 languages, fit seed 42). –: no concept feature selected, so no edit (score 0); *: a gate failed on test (ungated, on the original translations: Pornography A3z 0.76, Artificial intelligence A2 0.48 and A3z 0.52). A3z was defined after this run.

Test	n	Δ [95% CI]	W–L	p
H1: A2 − A1	2	$-.03$ [$-.29$, $+.23$]	1–1	1.00
H2: A3z − A3	10	$+.19$ [$+.03$, $+.35$]	7–3	.064
ungated	10	$+.26$ [$+.13$, $+.38$]	8–2	.014
H3: A3z − A2	2	$+.11$ [$-.11$, $+.32$]	1–1	1.00
A3z − A2, either	10	$+.34$ [$+.15$, $+.53$]	8–1	.012
A3z − A1, either	14	$+.07$ [$-.14$, $+.26$]	9–4	.34
Concepts edited (of 20): A1 12, A2 2, A3 10, A3z 10				
Mean strict: A1 .162, A2 .040, A3 .114, A3z .210				

Table 7: Pre-registered tests on the 20 held-out concepts (strict score unless noted; a concept without an edit scores 0). n: concepts in the test, i.e. those both approaches edit (H1), that A3 edits (H2), that all four edit (H3), or that either edits (last two rows). Bootstrap CI over concepts; W–L: concepts won and lost; two-sided Wilcoxon signed-rank p.

Coverage across fit seeds We repeated selection, without evaluation, with fit seeds 1 and 2 (Table 8). The counts are stable, and the gap between A2 and CINDER's selection holds at every seed and in both concept sets; but which concepts are edited changes with the seed, so that the union over three seeds covers 31 (A1), 13 (A2) and 29 (A3) of the 38 concepts.

	Original 18	New 20	All 38
A1	14/13/15 (16)	12/11/13 (15)	26/24/28 (31)
A2	8/6/7 (8)	2/3/5 (5)	10/9/12 (13)
A3, A3z	11/9/12 (15)	10/8/8 (14)	21/17/20 (29)

Table 8: Concepts with an edit for fit seeds 42 / 1 / 2 (union over the three seeds in parentheses). A3z uses A3's selection.

Seed-union follow-up Defined as an amendment to the pre-registration after the coverage study and before any of its evaluations, the seed union pools the three seeds' edits for both A1 (A1U) and A3z (A3zU), so that neither gets extra draws alone: every token that some seed edits is edited, with the mean of the seeds' edit vectors; all other settings are frozen (52 units newly evaluated; Table 9). Pooling helps A3z reliably (A3zU − A3z $+0.082$, 95% CI $[+0.043, +0.128]$, 19–4, $p < 0.001$; on the new 20 alone $+0.068$, 8–2) and A1 not ($+0.054$ $[-0.033, +0.151]$, 14–11, $p = 0.58$): A1U gains concepts, such as Tuberculosis ($0 \to 0.93$) and The Matrix ($0 \to 1.00$), but loses on concepts it already erased, such as Harry Potter ($0.79 \to 0.38$) and Cricket ($0.56 \to 0.27$), likely because its union of 8 languages $\times$ 3 seeds touches many more shared pieces, so the gates fire at a smaller δ. The pooled approaches tie (A3zU − A1U $+0.011$ $[-0.114, +0.137]$, 19–14, $p = 0.75$), which misses the pre-registered non-inferiority criterion (CI lower bound above -0.05); where both edit, A3zU leads by $+0.082$ (16–10, $p = 0.32$). Pooling also reaches Hindi in one case: Tuberculosis A1U erases it ($H_L = 0.77$) through features found only at seeds 1 and 2.

All 38 concepts	A1	A3z	A1U	A3zU
Concepts edited	26	21	31	29
Mean, strict	.248	.231	.302	.313
Mean, ungated	.252	.282	.320	.349

Table 9: Seed-union follow-up: single-seed (fit seed 42) and pooled (seeds 42, 1, 2) per-language EMBER and CINDER-z; a concept without an edit scores 0.

Language identity in the single fit Each token of the multilingual $\mathcal{V}^*$ is assigned to a language when at least 80% of its occurrences, normalized by each language's token count, come from that language. In the raw sample matrix, the language means carry 1.6% of the centred energy against 0.047% for the concept-vs-neutral mean (median ratio 34×), the concept is 2.9× quieter than in an English-only matrix (38 of 38 concepts), and the top principal directions are explained by language ($\eta^2 = 0.25$) rather than by the concept (0.004). Whitening against the eight-language background halves the language share (1.6% → 0.8%) and cuts the enrichment of the top contrastive directions from 8.2 (English background) to 2.6, but those directions remain single-language clusters. 8.6% of the raw fit's features and 13.7% of its judged candidates are dominated by one language (mostly German and English function words), against 4.9% and 0% for the sharpened fit. The raw fit nevertheless contains a lexicon-matching concept feature in 25–27 of 38 concepts per seed (sharpened: 23–28); the token judge accepts candidates with 3–4 lexicon words far less often when they come from the raw fit (5% vs. 30%), e.g. describing Cannabis's *marijuana, cannabis, alcohol, tobacco, THC* as "psychoactive substances". Blind labels of the features accepted only by the lexicon path are specific or broader, never unrelated, for both fits.

Coverage by fit and judge Table 10 re-judges the saved fits. With EMBER's judge, sharpening raises coverage from 10 to 16 concepts (seed 42); with the two-way judge, the raw fit reaches 18 and the sharpened fit 21, a difference that is not signif-

Concepts covered (of 38)	EMBER judge	two-way judge
Raw fit	10 / 9 / 12	18 / 20 / 18
Sharpened fit (CS+GS)	16 / 16 / 16	21 / 17 / 20

Table 10: Concepts with an accepted feature in the multilingual fit (Llama-3.1-8B, seeds 42 / 1 / 2), by fit and judge. Off-diagonal cells re-judge the saved fits and are counterfactual: no edit was evaluated for them.

icant over three seeds, so either half of CINDER's selection closes most of the gap. CINDER's selected features also spread over more scripts (34% use two or more, against 15% for per-language EMBER and none for the multilingual EMBER fit), although much of their Devanagari and Thai mass is generic words.

Selection cost Every fit's judge sees its top 10 candidates by ρ, with two calls each (§D). Per concept (fit seed 42), per-language EMBER runs 8 factorizations over about 30,600 rows with 160 judge calls (USD 0.015); the single fits run one factorization over about 6,500 sampled rows plus a coefficient solve for the full set, with 20 judge calls (EMBER: USD 0.0019) or 20 calls and one lexicon call (CINDER: USD 0.0047, 0.0019 once the lexicon is cached across seeds). The lexicon call is long, 20 words in each of the eight languages, and costs about as much as 30 judge calls. Factorization work (rows $\times$ iterations) drops about fivefold, but CINDER's wall-clock fit time (40 s on a GPU) matches EMBER's eight fits (44 s) because of the sharpening transform. Evaluation takes 2.5–3.5 GPU-hours per concept and approach, about 250 times the fit stage, and is the same for every approach.

Why Hindi and Thai resist EMBER edits only concept-only tokens, which never occur in the neutral sentences. Llama splits Hindi and Thai words into short pieces that also occur in everyday text, so almost none qualify: on Baseball, 32% of English question tokens are concept-only, against 13% in German, 6% in Thai and 5% in Hindi, and the share of questions containing an edited token falls from 0.94 (English) to 0.00 (Hindi). The same mechanism lets an edit made for one language act in every language that shares its pieces: Golf's per-language fit found its only feature in Thai, whose tokens cover 76% of the English questions.

Robustness Counts of concepts with an edit are stable across three fit seeds (A2: 9–12; A3: 17–21 of 38), but which concepts are edited is not, and the edited token sets of one concept overlap by a Jaccard index of only $\approx$0.2 across seeds, for every approach. In 6 runs, across all approaches, a δ that passed both gates on validation failed one on test.

G Collateral-Aware Evaluation: Details

G.1 Collateral Questions

For every edited token with a common meaning outside the concept (e.g., *Harry*, *Caesar*, *club*), LLM agents (Claude) wrote four-option multiple-choice questions that use the token in that meaning, and other agents reviewed them, both without knowing which methods would be scored. Items that the unedited model answers incorrectly under either option order were dropped, and each split was frozen with a hash before any edited model was scored. The held-out set has 3,501 items on 1,026 tokens. Collateral accuracy is normalized against the unedited model per token, like EMBER's metrics.

G.2 Scores, Gates and Selection Rules

H_3 is the harmonic mean of efficacy, specificity and collateral; H_4 adds Alpaca coherence. As all questions are fixed in advance, scoring a grid point needs only forward passes, whereas EMBER's coherence term calls an LLM judge on 150 Alpaca outputs per point. A point is admissible only if MMLU stays at or above 0.9 and validation collateral at or above 0.8 of the unedited model. Each rule selects one grid point per run on validation and scores it once on test:

Rule	Selects	Role
Collateral	max H_3, both gates	primary
Paper	max HM(efficacy, specificity), MMLU gate	the old protocol
Fixed	the point $r = 0.5$	equal edit size
Integrity	collateral rule, copy ≥ 0.9	words stay readable

G.3 Word Integrity Tests

Test	Question asked
Copy	Can the model repeat a sentence containing the word?
What is X?	Free answer, labelled blindly as knowing the concept (leak), knowing another meaning or not knowing (graceful), inventing a meaning (confabulation), or not registering the word (destruction)
Context	Is an unrelated MMLU question placed after a sentence with the word still answered?
Generation	Do 20 neutral prompts per concept that use the edited words keep their answer?

Free-text answers were labelled by LLM agents that never saw which method produced them; 10% of items were duplicated across agents, with agreement 0.92–0.97 on the exact label and 0.97–1.00 on the decisive class. Each test is also run on the real word, the word deleted, and a nonce. The *nonce floor* and the *deletion floor* are the efficacies of the unedited model on the concept questions with the edited words replaced by their nonces or deleted.

G.4 Pre-registration and Robustness

Analysis plans were written into a pre-registration and amendments, nearly all before their results existed (amendment 2, the intention-to-treat analysis and the McNemar test are exceptions). The primary endpoint pairs only runs where both methods edit; Table 11 varies this and the collateral gate (all post hoc, defined after the primary result).

Development claims on held-out concepts The evaluation was developed on 3 concepts (Harry Potter, Ancient Rome, Valentine's Day; 8 concept-seeds) and then frozen. Table 12 lists the development claims and what the 12 held-out concepts (60 runs) showed. Of the eight claims, four replicate, one replicates only in part, and three do not; the development ordering of the variants, in particular, did not hold, so the main text compares methods only on held-out concepts.

Mean and Noise (development only) Following EMBER, Mean replaces each edited embedding by the vocabulary mean and Noise subtracts a random direction of the same length; both use the edited tokens of EMBER or CINDER and the same grid, gates and selection (Table 13). Mean has a single point ($r \approx 0.85$–0.95), whose validation collateral is 0.71 (0.55–0.78), so it fails

	ΔH_3 vs. EMBER			
Setting	CINDER	-RS	-z	-RS-z
Primary	$+.042$	$+.034$	$+.132$	$+.084$
Holm p	.41	.41	.074	.074
All 60 runs	$+.123$	$+.133$	$+.167$	$+.178$
Holm p	.10	.10	.10	.049
Gate 0.9	$+.174$	$+.147$	$+.195$	$+.102$
Holm p	.012	.012	.008	.014
Gate 0.7	$-.019$	$-.041$	$+.121$	$+.070$
Holm p	.73	.73	.27	.72
Leave one out	.03–.06	.01–.06	.11–.16	.06–.10
p	.08–.38	.16–.70	.004–.04	.004–.05

Table 11: Robustness of the primary comparison (paired test H_3, unit = concept; 11 held-out concepts where both edit, 12 for all runs). Columns: CINDER, CINDER-RS, CINDER-z, CINDER-RS-z. Primary: runs where both edit. All rows below it are post hoc. All 60 runs: a run without an edit is scored as the unedited model. Gate: the collateral rule rerun on the saved grids with the collateral gate at 0.9 or 0.7 instead of 0.8. Leave one out: range of the mean difference and of the uncorrected p over the 11 omitted concepts.

the collateral gate in every run and is never selected; at that point it deletes the words (3 labelled runs: copy 0, destruction 0.99) while the rest of the prompt is still read (context accuracy 0.52–0.54). Noise at its selected points (median $r = 1$; same 3 runs) destroys 0.25–0.35 of the edited words and still leaks the concept in 0.29 of the free answers. We did not run the baselines on the held-out concepts.

G.5 Integrity-Gated Selection

Figure 6 shows the trade-off that this rule cuts through. The integrity-gated rule (post hoc) is the collateral rule restricted to grid points where at least 90% of the edited words can still be copied; the copy test was run at every grid point up to $r = 2$ (1,530 points). It coincides with the collateral-rule point in only 16 of 255 run-variants. Table 14 gives the result. On the paired endpoint of the primary analysis (runs where both edit), CINDER gains $+0.096$ H_3 over EMBER (95% CI $[-0.006, +0.214]$, 7–3 concepts, Holm $p = 0.42$) and CINDER-RS $+0.092$ (7–3, 0.42), the -z variants $+0.029$ and $+0.026$ (0.75). Counting runs without an edit as no erasure, CINDER gains $+0.134$ (10–2, Holm $p = 0.015$) and CINDER-RS $+0.166$ (11–1, 0.006), the -z variants $+0.053$ and $+0.063$ (0.33). The -z edits break copying at smaller r, so the rule pushes them to $r = 0.05$ in 24 of 52 and 26 of 55 runs. Alpaca coherence

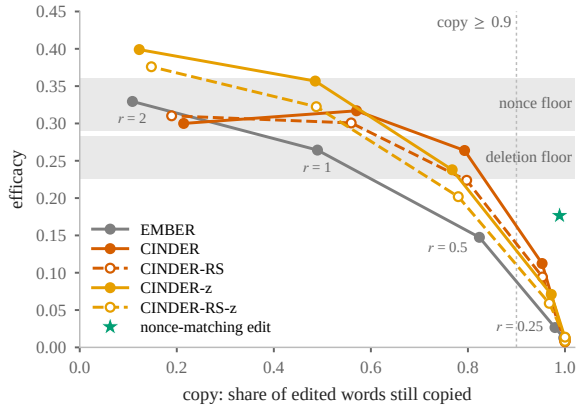


Figure 6: Removing more costs the words. Mean efficacy against the share of edited words the model can still copy, along the edit-size grid $r = 0.05, \ldots, 2$ (labelled on EMBER's curve), over the held-out runs where each method edits (41–55 of 60; validation questions). Grey bands: the efficacy of the unedited model when the edited words are replaced by their nonces (nonce floor, 0.29–0.36) or deleted (deletion floor, 0.23–0.28) in the test questions, over the five methods' token sets; dotted line: the integrity gate (copy ≥ 0.9). Every method crosses its (test) deletion floor between $r = 0.5$ and 1, where copy falls from about 0.8 to 0.5, and none reaches its nonce floor while more than half of its words remain copyable. Star: the free nonce-matching edit on CINDER's tokens (post hoc; seed 42, 9 runs, test split), which keeps 0.99 of the words copyable at efficacy 0.18.

at these points (seed 42) is 0.97–0.98 for every method.

G.6 Direction or Token Set

Four cells cross the token set with the direction at a fixed $r = 0.5$, on the runs where EMBER has a token set; "CINDER's tokens" are CINDER's top concept-only tokens, as many as EMBER edits. All cells use dense coefficients so that every token has a coefficient on $\mathcal{F}_C$.

Efficacy / H_3	EMBER direction	CINDER direction
EMBER tokens	.20 / .39	.26 / .46
CINDER tokens	.20 / .39	.26 / .48

The direction effect (mean of the row differences) is $+0.073$ efficacy and $+0.097$ H_3, on 10 of 11 concepts (Holm p 0.097 and 0.055); the token-set effect is $+0.016$ and $+0.024$ (n.s.); the direction costs 0.02 collateral. At $r = 1$ (secondary), the cell means are .27 / .47 (EMBER tokens, EMBER direction), .27 / .45 (EMBER tokens, CINDER direction), .28 / .46 (CINDER tokens, EMBER direction) and .29 / .48 (CINDER tokens, CINDER direction); the direction effect is -0.016 efficacy

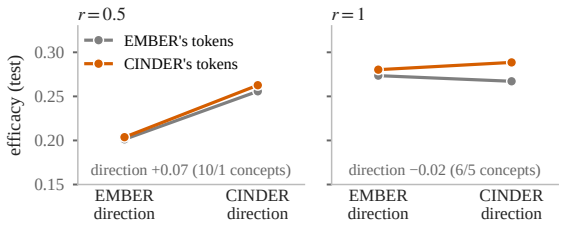


Figure 7: Test efficacy of the four direction × token-set cells (post hoc). At $r = 0.5$, switching to CINDER's direction adds $+0.07$ (10 of 11 concepts) and switching to CINDER's tokens $+0.02$, so the two lines nearly coincide; at $r = 1$ the direction effect vanishes (-0.02, 6/5 concepts). EMBER's own cell averages its 41 runs, the other cells 38.

($p = 0.97$) and -0.007 H_3 ($p = 1.0$), and the token-set effect $+0.010$ and $+0.012$ (both n.s.).

G.7 The Nonce-Matching Edit

We keep the model frozen and learn a change to each edited row. The loss pushes the edited model, on concept sentences and new definitional prompts, toward what the unedited model does when the word is replaced by its nonce, while two further terms keep the word copyable (new carrier sentences) and keep the other meanings of multi-meaning tokens (sentences the model generated itself); no evaluation item is used for training. The edit is either confined to the span of the edit set's concept components (rank ≤ 3) or free, and is learned on CINDER's or EMBER's token set; one free variant also adds concept questions that the model wrote itself to the forget data (+QA). Selection is the integrity-gated rule; a run reaches the *nonce point* when its test efficacy is at least its nonce floor minus 0.05, with copy ≥ 0.9 and collateral ≥ 0.8 (all post hoc, seed 42; Table 15). The nonce floor averages 0.36 on the runs with CINDER's tokens and 0.26–0.27 on those with EMBER's, and almost all runs that reach the nonce point have a floor near 0 (COVID-19 0.00, Pornography 0.07–0.11); only Baseball (free, EMBER's tokens: 0.36 vs. floor 0.31) and Heroin (+QA: 0.44 vs. 0.49) reach a substantial floor. Free edits move the rows far (median $r = 3$, 3.2 and 1.6) and keep them copyable, and remove 2–3× as much as integrity-gated EMBER and 1.3–1.6× as much as integrity-gated CINDER on the same runs; paired H_3 against it (uncorrected): free (CINDER's tokens) − CINDER-z $+0.20$ (7 of 9 concepts, $p = 0.055$), +QA − CINDER-z $+0.26$

(7–1, $p = 0.039$), free (EMBER's tokens) − EMBER +0.28 (6–1, $p = 0.047$). But asked "What is X?", the free edits repeat the word in only 0.34–0.49 of the answers (unedited word 0.96, nonce 0.75) and name a different invented string in 32–47% (nonce 17%), which the blind labels count as destruction. Inside the concept subspace the words stay intact but almost nothing is removed, and the concept still leaks in 0.39 (CINDER) and 0.53 (EMBER) of the answers.

H Embedding-Space Analyses: Details

All analyses in this section use the saved factorizations of 38 concepts × 5 seeds on Gemma-2-2B-it: EMBER's matrix with row sparsity (EMBER), the sharpened matrix with row sparsity (CS), and the sharpened matrix with global sparsity (CS+GS). Unless stated otherwise, the concept feature of a run is identified without the judge, as the feature whose words match the concept's lexicon most often.

H.1 Where the Concept's Energy Lies

Matrices and fits Raw $E^{\top}_{\mathcal{V}^*}$; raw with $\hat{\mu}_B$ projected out; centred at μ_B; whitened only ($\gamma = 0$); and CS. Row-sparse fits, 38 concepts × 5 seeds; the 570 fits for the three intermediate matrices are new. A run "finds" the concept when its best feature has at least 5 lexicon words among its first 20 source words; no LLM judge is used.

Frequency Over 6.2M tokens of English Wikipedia, concept-only tokens are rarer than neutral-only ones in 35 of 38 concepts (median 9.4 vs. 12.9 per million), but their norms are not smaller (raw energy 0.9% larger). Across the vocabulary, a token's projection on $\hat{\mu}_B$ falls monotonically with its frequency (Spearman -0.69 within a concept), and rare tokens put a smaller share of their centred energy on the top background directions (0.07 vs. 0.65 at the frequent end). The tokens a raw concept feature collects are the rarest in $\mathcal{V}^*$ (median 2.5 vs. 23 per million).

Seeded feature and full fits The seeded single feature (Figure 2b) ends with purity 0.24 in the raw matrix, 0.04 centred and 0.27 with $\hat{\mu}_B$ projected out; whitening makes it pure (0.86, keeping 41% of the lexicon tokens) but ordinary (70th percentile of feature values), and sharpening raises its value to the 91st percentile at the same purity (37 of 38 concepts). In full fits, a lexicon concept feature is found in 56% of raw runs, 61% with $\hat{\mu}_B$ projected out, 59% centred, 49% whitened only and 69% with CS; per concept, raw → CS is 15 better and 3 worse ($p = 0.011$), and whitened only → CS 17 and 0. The raw fit's best lexicon feature usually ranks near the top by ρ (median rank 2, against 1 under CS; $p = 0.92$); Harry Potter's raw rank is seed-dependent (12, 1, 28, 48, 1), while CS ranks it first in all five seeds.

Selection-free directions The concept-minus-neutral mean direction, the top generalized eigenvector of concept vs. neutral second moments, and the top direction of the lexicon tokens do not rise in energy rank with CS (e.g., lexicon direction rank 2 raw, which reflects $\hat{\mu}_B$; 9 without $\hat{\mu}_B$; 16.5 with CS). What sharpening changes is the energy of the lexicon direction relative to whitening alone (3.1× vs. 11.4× an average direction) and the value of the concept's cluster (App. B).

H.2 Feature Geometry and Norms

Setup 570 saved fits (EMBER, CS with row sparsity, CINDER), directions sign-aligned and compared in four metrics: the fit's own space, the raw embedding space (sharpened directions mapped back), the sharpened space (raw directions mapped forward), and the raw space with $\hat{\mu}_B$ projected out of every direction. Semantic relatedness of two features is the cosine of a small sentence encoder (all-MiniLM-L6-v2) between their source-word lists, independent of Gemma; token frequencies come from 32.6M tokens of Wikipedia.

Orthogonality Mean $|\cos|$ between directions:

Metric	EMBER	CS+RS	CINDER
Own fit space	.164	.052	.051
Raw embedding space	.164	.291	.258
Raw, $\hat{\mu}_B$ projected out	.079	.086	.079

Mapping sharpened directions back multiplies them by $T^{-1} \propto \Sigma_B^{1/2}$, which re-inflates the dominant background direction; $\hat{\mu}_B$ has cosine 0.82 with the top background principal component.

Neighbours and meaning The concept feature's nearest-neighbour angle is 50.6° (EMBER), 73.6° (CS) and 74.3° (CINDER); EMBER's neighbour is a recurring feature of rare names from the neutral sentences (*Fulda*, *Amiens*, *Ferdinando*) at the 59th relatedness percentile, the sharpened neighbours at the 92–94th. Over all pairs, the rank

correlation of cosine with relatedness is higher for EMBER (0.33 vs. 0.24), but EMBER's raw cosines largely track word frequency; controlling for it, CS is higher (0.34 vs. 0.25), and among the top 0.5% of pairs by cosine only the sharpened dictionaries are semantic (75–79th vs. 51st percentile). Mapped back to the raw space, sharpened cosines lose this property: it belongs to the metric in which they were found. Projecting $\hat{\mu}_B$ out of EMBER's directions alone opens its nearest-neighbour angle from 50.6° to 74.0° (38 of 38 concepts), the sharpened level.

Norms The concept feature's norm is at the 82nd percentile under EMBER and the 57–61st under CS; 45% of its squared norm lies along $\hat{\mu}_B$ (typical feature 12%; 42% with the identification of App. H.1), as does 97% of its excess squared norm over the run mean, and with $\hat{\mu}_B$ removed it falls to the 63rd. A token's coordinate along μ_B falls from 1.1 (rarest decile) to 0 (most frequent), so features built from rare tokens with same-sign coefficients collect about $\sqrt{w}\,\|\mu_B\|$ of norm. For Harry Potter, the raw feature is the 3rd to 6th largest of 100 by $\|\mathbf{z}_i\|$, with 0.67–0.80 of its squared norm along $\hat{\mu}_B$. Neither norm measures concept strength: with unit-norm coefficient rows, $\|\mathbf{z}_i\|$ is the scale of the support's vectors in the fit's metric.

Splits Two or more lexicon-matching features occur in 4–12% of runs (15–28% by the judge). Blind labels of the 56 split pairs give 52 distinct sub-topics, 2 cuts of one concept and 2 off-topic pairs (e.g., batting vs. pitching, card games vs. betting). Ancient Rome has a Rome feature and an empire feature in all 15 runs; under GS the Rome feature grows to 92–109 tokens and absorbs Roman persons, whereas EMBER's fixed 48 slots pad it with other capitals (Milan, Paris, Ottawa).

H.3 Feature Widths and Tails

See App. C.1.

H.4 Abstract Concepts and Contamination

Abstract concepts 13 concepts that sit between broader topics (Romanticism, Utilitarianism, Baroque, Gothic architecture, Surrealism, Art Nouveau, Neoclassicism, Gnosticism, Existentialism at $|\mathcal{S}_C| = 300$; Stoicism, Epicureanism, Impressionism, Neoplatonism, whose pages are shorter, at $|\mathcal{S}_C| = 190$) and 4 entity controls (Harry Potter, Baseball, Uranium, Golf, at both sizes), 5 seeds, EMBER, CS and CS+GS (615 fits). Each of the top 10 features by ρ was shown, blind to condition, as its 20 strongest tokens with source words and labelled specific (P), broader (B) or unrelated (N) by two independent LLM annotators (agreement 0.94, $\kappa = 0.88$; parent topics fixed before any fit). At $|\mathcal{S}_C| = 300$, a P feature is among the top 3 in 7% of abstract and 50% of entity runs under EMBER (Mann–Whitney $p = 0.043$), and CS changes top-10 presence by $+0.04$ (n.s.); CS+GS gains $+0.22$ from two concepts (Baroque, Utilitarianism). Stoicism illustrates the pattern: its top features are emotions, philosophy and religion, ethics and discourse words, and its most Stoic feature is a spelling feature that mixes *Stoic* and *stoa* with *stock* and *store*.

Contamination For Harry Potter, Baseball and Ancient Rome, a fraction $f \in \{0.1, 0.25, 0.5\}$ of the 300 concept sentences is replaced by sentences from adjacent pages (Harry Potter: Magic, The Lord of the Rings, The Chronicles of Narnia, Fantasy literature; Baseball: Softball, Cricket; Ancient Rome: Ancient Greece, Byzantine Empire), or, as a control added afterwards, from unrelated pages. Sentences naming the concept are dropped, and the sharpening target stays the concept's own page. EMBER's mean rank of the best P feature rises from 4.1 to 6.9 at $f = 0.5$, equally with unrelated sentences (5.9 vs. 5.7, $p = 0.96$), so the loss reflects fewer concept sentences rather than a fading contrast; for $f > 0$, CS keeps it in the top 10 in 80–93% of runs, at a mean rank of 1.8–3.3. The share of the feature's top tokens that come only from the inserted sentences grows with f (EMBER 0.14 / 0.20 / 0.25; CS 0.05 / 0.13 / 0.14); sharpening toward the contaminated sentences instead of the clean page raises it to 0.23 at $f = 0.5$. At $f = 0.25$, EMBER's Harry Potter feature mixes *Tolkien*, *Gandalf* and *Hobbit* into the house and character names, whereas CS's at $f = 0.5$ keeps only Tolkien and Gandalf, at positions 16–20.

H.5 Extrapolation Beyond the Sample

Benchmark 17 concepts with at least 80 page sentences outside the 300 sampled ones and a concept feature for both EMBER and CINDER in some seed; on average 734 new concept-only token types per concept, plus new neutral token types from a 1,000-page Wikipedia sample (balanced: as many new neutral as new concept-only

tokens; natural: one neutral sentence per new page sentence). Three saved dictionaries, kept fixed (EMBER; CS with row sparsity; CINDER), 5 seeds. Ground truth: blind labels as concept-specific (S), adjacent (A) or unrelated (U) on 5,196 tokens, written by LLM agents that saw only the concept, the token, its source words and a short context; agreement on 300 double-labelled items 0.90 ($\kappa = 0.82$).

Rules Scores: projection on the concept feature, cosine, the coefficient of the joint dense least-squares solve, and the code share $y_{ft}|y_{ft}|/\|\mathbf{y}_{\cdot t}\|^2$. Admission: the fit's own sparsity rule over the extended set (1%), the saved absolute width, an in-sample cut-off ($|y_{ft}|$ at least the weakest coefficient of the saved support), the token's dominant feature, and share ≥ 0.5.

Results Top-10 precision among new concept-only tokens is 0.18–0.27 for every dictionary and score (random 0.065); counting adjacent tokens it is 0.49–0.69, and no score separates concept-specific from adjacent tokens (AUC 0.40–0.48). The 1% re-solve admits 15–22 tokens per run at precision 0.22–0.25, about 37% of them unrelated; fixed width and cut-off give the same precision, but fixing the width pushes out 13–15% of the original support, and Harry Potter's width grows from 48 to 59 under the 1% rule. Routing a token's shared meaning to other features (dominant feature or code share) does not help. On EMBER's raw dictionary, projection on the concept feature mostly ranks rarity (Spearman 0.70 with Wikipedia rarity, against 0.25 for the joint solve); in the sharpened space the concept feature is nearly orthogonal to the rest of the dictionary (largest $|\cos|$ 0.28–0.30 vs. 0.64), and projection and joint solve agree. Recall is low: on six fully labelled concepts, re-solving the sparse codes recovers 4% of the concept-specific unsampled tokens. CINDER's wider features admit more concept-specific tokens per run at equal precision (6.1 vs. 3.9), and the sharpened dictionaries rank lexicon words higher (AUC 0.91 vs. 0.67 with projection). 87% of the unsampled page tokens occur once on the page (52% of concept-only $\mathcal{V}^*$ tokens), and precision rises with a token's page frequency, from 0.28–0.30 at one occurrence to 0.59–0.71 at ten or more (in-sample); at one occurrence, in-sample and extrapolated tokens are alike. We recommend ranking by the joint solve, admitting with the in-sample cut-off, and preferring CINDER's dictionary. Figure 8 shows both results.

I Qualitative Examples

The Harry Potter feature, before and after sharpening Table 16 lists the 48 tokens of the Harry Potter feature (1% row sparsity, seed 42) on EMBER's raw matrix and on the sharpened matrix, by decreasing coefficient, written as the words they come from. It is taken from our exploratory notebook fits, which differ slightly from the saved fits of Figure 9 (there, ranks 20–30 of the raw feature hold 3 Harry Potter tokens, and the sharpened feature has no neutral token). Both features open with 19 unmistakable Harry Potter tokens. After that, the raw feature drifts into unrelated names, places and taxonomic suffixes, while the sharpened feature continues into the fictional world (Gringotts, Hogsmeade, Quidditch, McGonagall, Hagrid) and is more consistent across fragments of the same word (*Gry*, *Hog*, *warts*).

Free text at the selected strength Table 3 shows two prompts whose edited words are used outside the concept. A third, from the same run: asked "In which year was Donald **Trump** first elected President [...]?", the model answers "[...] in 2016" at $\delta = 1$ and only "Donald Trump" at $\delta = 200$.

Judge rejections EMBER's raw fits fail in the same way: its Heroin feature (seed 42) is described as "substances, medications, and illicit drugs along with related concepts like pharmacology and addiction" and its World War II feature (seed 43) as "20th-century totalitarian political ideologies, regimes, and associated historical atrocities", and both are ruled not to be the concept (confidence 0.85); these are the descriptions quoted in §3. Table 17 shows three more concept features that EMBER's judge (TJ) rejected, from the sharpened fits with EMBER's 1% row sparsity. In each case the top tokens are plainly the concept, the stage-1 description, written without seeing the concept, widens it to a broader category (even when, as for Valentine's Day, it names the concept), and stage 2 answers that the feature is *not* the concept, with confidence 0.85. Because the answer is "not a member", these rejections do not depend on the confidence threshold.

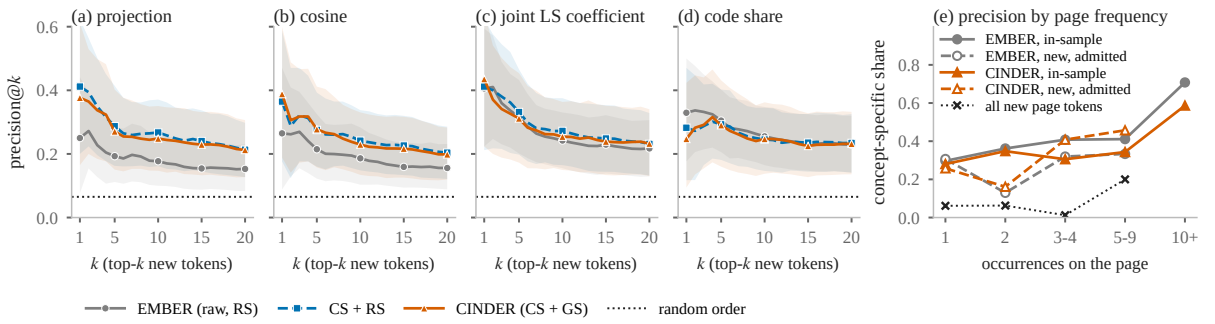


Figure 8: Extrapolating a fixed dictionary to concept-only tokens outside the 300 sampled sentences (17 concepts, 5 seeds, balanced condition). (a)–(d) Precision@k: share of the top-k new concept-only tokens by each score that are blindly labelled concept-specific, mean over concepts with 95% bootstrap bands; dotted: random order (0.065). (e) Concept-specific share by the token's number of occurrences on the concept's page, for the tokens of the in-sample support (solid) and the new tokens admitted by the 1% re-solve (dashed); dotted: all new page tokens. Bins with fewer than 5 concepts are omitted.

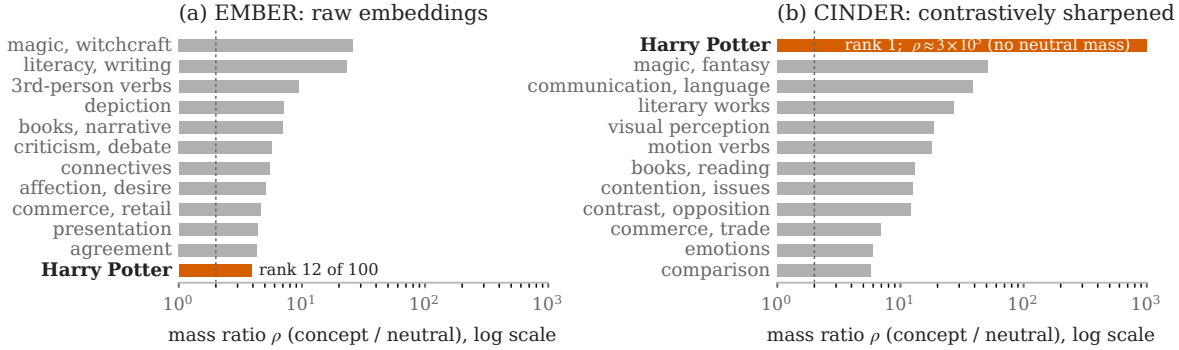


Figure 9: The top 12 of 100 sparse-MF features for Harry Potter on Gemma-2-2B-it at seed 42, ranked by EMBER's mass ratio ρ (Eq. 1); labels paraphrase each feature's LLM description. (a) Raw embeddings: eleven generic topics outrank the concept feature. (b) Sharpened matrix, same 1% row sparsity: the concept feature ranks first and has no neutral token. The raw rank varies with the seed (12, 1, 28, 48, 1 at seeds 42–46), while sharpening ranks the concept first at all five; across concepts, the raw rank is usually already near the top (§6). The dotted line marks $\tau = 2$.

J Reproducibility and Cost

Code and results All code, pre-registrations and result files are in our repository.

- Single-language fits and features: `experiments/varwidth/` (fitting `core.py`, data `prep.py`, selection `select.py`); sharpening `experiments/whitening/sharpen.py`.
- Collateral-aware evaluation: `experiments/varwidth/erase_v2.py`, `collateral.py`, `integrity.py`, `last_mile.py`, `nonce_edit.py`, `confirm_stats.py`; pre-registration `results/varwidth/layers/PREREG.md`; results in `results/varwidth/layers/collateral/` (`confirm.md`, `last_mile.md`, `twobytwo.md`, `nonce_edit.md`).
- Feature-level ladder and judges: `results/varwidth/layers/SUMMARY.md`, `experiments/varwidth/j3_eval.py`.
- Multilingual study (branch `orip-ml`): `experiments/ml/`, `results/ml/PREREG_RESULTS.md`, `results/ml/UNION_RESULTS.md`.
- Embedding-space analyses: `experiments/paper_analysis/`, `results/paper_analysis/`.
- Figures: `paper/figures/src/`.

Compute and cost All GPU work ran on the university's SLURM cluster, on NVIDIA TITAN Xp and GeForce RTX 2080 Ti nodes, one GPU per job for the jobs of Table 18 (one used two); factorizations of the single-language study ran on CPU nodes. Table 18 gives the GPU time allo-

cated to each study and its Gemini spend (Gemini 3.5 Flash-Lite). In the multilingual study, evaluation dominates: 2.5–3.5 GPU-hours per concept and approach, 68, 34, 71 and 66 GPU-hours for A1, A2, A3 and A3z at fit seed 42, against under a minute per concept for the fits; its seed-42 fits and about 150 GPU-hours of evaluation (50 units) ran on workstation GPUs that the accounting does not record. The single-language feature-level ladder spent USD 2.03 on feature judging (TJ, WJ, lexicons and a 10% re-judge for noise) and USD 2.41 on Alpaca coherence for its erasure anchor; the collateral-aware evaluation used Gemini only for Alpaca coherence (USD 1.92 in development, 2.24 for the seed-42 last-mile check and 1.54 at the integrity-gated points), since its questions and all free-text labels were written by LLM agents (Claude) instead; the multilingual study spent USD 2.24 on selection (judge and lexicon calls), 2.16 on generating and verifying the questions of the 20 new concepts and 0.20 on the stability study. The spend is logged with deliberately conservative prices (USD 0.30 or 0.25 per million input and 2.50 per million output tokens), so it is an upper bound. SLURM accounting also shows about 540 GPU-hours on 3–8 October, mostly on RTX 2080 Ti nodes (446 for a second-pass δ search of the judge re-run), which belong to the preliminary EMBER/SNMF study of App. A but are partly recorded under job names that we cannot attribute to a single study; they are not in Table 18.

Development claim	Held-out result	Verdict
Without collateral, selection picks destructive edits (EMBER at $r = 10$–100 in 6 of 6 runs)	paper rule: EMBER and CINDER-z at median $r = 100$; ours: every method at median $r = 1$	replicated
At small r, CINDER's direction removes more (efficacy .33 vs. .20 at $r = 0.5$, validation)	at $r = 0.5$, every variant beats EMBER on 10 of 11 concepts (CINDER ΔH_3 $+.15$, Holm p .012)	replicated
The gain is the direction, not the token set (Δ_{dir} efficacy $+.15$ at $r = 0.25$, $+.09$ at $r = 1$; 5 runs)	$+.07$ at $r = 0.5$ (10 of 11, Holm p .097); $-.02$ at $r = 1$	at $r = 0.5$ only
At their own operating points the methods tie (test H_3 .42 vs. .42)	CINDER $+.04$, Holm p .41	replicated
CINDER-RS is the strongest variant (H_3 .46)	CINDER-RS $+.03$; the -z variants lead ($+.13$, $+.08$)	not replicated
The original-space edit (-z) does not help ($-.01$ and $-.06$ vs. its parent)	CINDER-z $+.13$ vs. EMBER (9 of 11, Holm p .074)	not replicated
CINDER destroys fewer words (destruction .21 vs. .30; exclusive words .37 vs. .74)	.52 vs. .49; exclusive .70 vs. .65 (n.s.)	not replicated
Under the paper's rule EMBER beats CINDER, and the -z edit reaches EMBER by deleting the words	CINDER wins 1 of 11 concepts on H_2; CINDER-z wins 11 of 11 at $r = 100$	replicated

Table 12: Development claims (3 concepts, 2–3 seeds) and their held-out test (12 concepts, 5 seeds, test split). Differences are paired against EMBER on runs where both edit, unless stated otherwise.

Method (tokens)	Runs	Eff.	Coll.	H_3	mean r
EMBER	6	.22	.93	.42	0.76
Noise (EMBER's)	6	.22	.91	.40	1.09
Mean (EMBER's)	6	no admissible point			
CINDER	7	.21	.87	.42	1.71
Noise (CINDER's)	7	.18	.89	.34	1.57
Mean (CINDER's)	7	no admissible point			

Table 13: Embedding-editing baselines on the development concepts (test split, collateral rule; mean over concept-seeds). A run without an admissible point is not edited.

Method	Runs	Eff.	Coll.	H_3	r	Copy ex./mu.	H_4
EMBER	41	.06	.98	.12	0.1	.98/.99	.18
CINDER	52	.12	.97	.25	0.25	.96/.97	.35
CINDER-RS	55	.14	.98	.27	0.25	.94/.97	.20
CINDER-z	52	.07	.99	.16	0.25	.97/.99	.16
CINDER-RS-z	55	.07	.99	.16	0.1	.98/.98	.25

Table 14: Integrity-gated selection (copy ≥ 0.9; post hoc) on the 12 held-out concepts, test split. Runs: runs with an edit (of 60); means over those runs; r: median edit size; copy for concept-exclusive and multi-meaning words separately; H_4 adds Alpaca coherence (seed 42 only, 8–10 runs).

Edit (tokens)	Nonce point	Eff.	Coll.	Copy	Destr.	Broken
Learned, inside the concept subspace						
CINDER's	1/10	.04	.99	.99	.01	.02
EMBER's	2/8	.02	.96	.97	.04	.04
Learned, free						
CINDER's	2/9	.18	.86	.99	.50	.16
EMBER's	3/7	.21	.86	.99	.59	.15
CINDER's +QA	3/10	.22	.89	1.00	.47	.15
Subtraction, integrity-gated (same runs)						
EMBER	–	.07	.98	–	–	–
CINDER	–	.14	.97	–	–	–
CINDER-z	–	.06	.99	–	–	–

Table 15: The nonce-matching edit (post hoc; seed 42, 12 held-out concepts, test split). Nonce point: runs that reach the nonce point, of the runs with an edit; Copy: share of edited words copied; Destr.: share of "What is X?" answers labelled as not registering the word; Broken: share of the 20 neutral prompts per concept whose answer breaks.

Ranks	EMBER (raw)	CINDER (sharpened)
1–19	all Harry Potter (Gryffindor, Dumbledore, Slytherin, Rowling, Voldemort, Weasley, …)	all Harry Potter (Gryffindor, Slytherin, Dumbledore, Weasley, Hogwarts, …)
20–30	Kipling, **Ginny**, **Lupin**, Beatles, Arrondissement, Escondido, vice-chancellor, Cerambycidae, Alamos, Sherlock, Balompié	**Hogwarts**, **Gryffindor**, **Ginny**, **Hogsmeade**, **Gringotts**, **Hogwarts**, **Lupin**, **Remus**, **Granger**, **Slytherin**, **JK**
31–48	**Remus**, Chrysomelidae, Trump, Nestlé, Bloomsbury, tesla, Fiennes, Polygyridae, Cerambycidae, Picasso, Assassins, Steelers, **Gryffindor**, Chhattisgarh, Lego, Kinect, Reagan, HBO	**Sirius**, **wand**, **McGonagall**, **Muggle**, **Ron**, **Goblet**, Sherlock, **Hagrid**, **Quidditch**, **Beauxbatons**, **potions**, **Rowling**, **Riddle**, **Grindelwald**, **Quirrell**, Holmes, **Merlin**, NHP

Table 16: Tokens of the Harry Potter feature in EMBER's raw matrix and in the sharpened matrix (seed 42), shown as their source words. Bold: clearly Harry Potter tokens beyond rank 19. Ranks 20–30 hold 2 of 11 Harry Potter tokens before sharpening and 11 of 11 after.

Concept	Top 20 tokens	Stage-1 description
Heroin (seed 42, $\rho = 95$)	drug(s), opium, opioid(s), narco, cocaine, heroin, opi, fentanyl, medications, morphine, pharmaceutical, pharmacology, pharmaco, addiction	"pharmacological substances, specifically focusing on narcotics, opioids, and addictive drugs"
World War II (seed 42, $\rho = 31$)	communist, Nazi(s), communism, fascist, Hitler, fascism, Soviet(s), socialist, Adolf, totalitarian, Mussolini, Germans, Holocaust	"20th-century totalitarian regimes and their associated political ideologies and figures"
Valentine's Day (seed 43, $\rho = 21$)	Valentine(s), Valentin, Christmas, Valent, Valentín, Halloween, Easter, holiday(s), Valentino, and the Chinese for "lover"	"the concept of celebratory holidays and festive occasions, with a particular emphasis on Valentine's Day"

Table 17: Concept features rejected by EMBER's judge (sharpened matrix, 1% row sparsity). Tokens are listed in the judge's order with case and spacing variants merged; each description is quoted from the stored stage-1 answer, and stage 2 answered `is_member = false`, confidence 0.85, for all three.

Study	GPU-h	GPU	Gemini (USD)
Feature ladder	20	TITAN Xp	4.43
Collateral evaluation	346	both	5.70
Multilingual	320^{+}	TITAN Xp	4.62
Analyses (§6)	11^{+}	TITAN Xp	0
Total	696^{+}		14.76

Table 18: Compute and LLM cost per study (17–23 October 2026). GPU-h: allocated GPU time from the cluster's accounting (182 TITAN Xp and 164 RTX 2080 Ti hours for the collateral study); $^{+}$: SLURM only, plus unrecorded workstation GPU time (about 150 GPU-hours for the multilingual evaluation; some analyses).